



DEPARTMENT OF ENGINEERING MATHEMATICS

Cell Line Selector: An Integrated Multi-Omics Recommendation System for Cell Line Selection

*An Evidence-Based Framework for Integrating Heterogeneous Biological
Datasets to Identify and Rank Suitable Cell Lines*

Chaithali Shashikanth Kundapur

Musa Khumalo

Thiruvél Andagurunathan Pandian

Triveni Dhamdhere

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requirements of the degree of Master of Data Science in the Faculty of
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Supervisors: Dr. Daniel D'Andrea

Declaration

This dissertation is submitted to the University of Bristol to fulfill the requirements for the MSc degree in the Faculty of Engineering. It has not been submitted for any other degree or diploma to any examining body. Except where specifically acknowledged, all work presented is that of the Authors.

This dissertation presents research conducted as a group project. The framework described in Chapter 3 was collaboratively designed and implemented. The analysis, evaluation, and written account presented are the joint work of the Authors listed below. Any component, dataset, or result originating outside this project is identified by specific reference at the point of description. All external data sources are public and acknowledged in Chapter 2 and in the references. The industrial brief motivating this work is acknowledged in §1.2.

Chaithali Shashikanth Kundapur

September 11, 2026

Musa Khumalo

September 11, 2026

Thiruvel Andagurunathan Pandian

September 11, 2026

Triveni Dhamdhere

September 11, 2026

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List of Technologies

The system components referenced throughout Chapters 3 and 4 are summarised in Table 1. The harmonisation and statistical stages depend only on pinned data releases and fixed parameters, so a rerun reproduces every score exactly; the agentic layer depends on a hosted model and two live nomenclature services, so its outputs are reproducible only to the extent those services are stable (§3.6).

Table 1: Technologies used in this thesis.

Component	Role	Version
Python	Ingestion, identifier resolution, evidence models, composite score, query layer	python:3.12-slim
DuckDB	Sole persisted artefact; harmonised corpus and score tables	1.5.5
pandas	Table loading, reshaping and joins	2.3.3
NumPy / SciPy	Robust estimators, inverse-normal transform, shrinkage, Monte-Carlo uncertainty layer	1.26.4 / 1.17.0
Amazon Bedrock	Model hosting	region:eu-west-2, version: -
Claude Haiku 4.5	Tool routing and explanatory prose; temperature 0.0, 2048 output tokens	modelID: eu.anthropic.claude-haiku-4-5-20251001-v1:0
LangChain AgentExecutor	Single-step tool selection	langchain==1.3.14, langchain-classic==1.0.8
AWS Lambda	Execution environment for the agent endpoint	python:3.12-slim
AWS API Gateway, IAM	Request routing and access control	—
REST APIs	External identifier resolution	Ensembl + HGNC on live services
FastAPI	Query endpoints for the three query modes of §3.5	fastapi==0.135.1, starlette==0.52.1, uvicorn==0.42.0

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Table 1 continued from previous page

Component	Role	Version
React, TypeScript	Single-page client and typed adapter layer	react==18.3.1, react- dom==18.3.1, type- script==5.9.3
Vite	Build tooling	vite==7.3.6, @vitejs/plugin- react==4.7.0
Tailwind CSS	Styling	tailwindcss==3.4.19, postcss==8.5.26, autopre- fixer==10.5.4
Lucide	Iconography	lucide- react==0.344.0
Git, GitHub	Version control; pull-request workflow with branch protection	—

Project Links

The source code and the deployed application supporting this dissertation are available at the addresses below.

Resource	Address
Source repository	https://github.com/UoB-AstraZeneca-25-26/UoB-GeneTraceAI-25-26
Deployed application	http://genetraceai-cellline-ui.s3-website.eu-north-1.amazonaws.com

Notation and Acronyms

The notation used throughout this thesis is summarised in Table 2. Definitions are given in the main text and relevant appendices. Two symbols carry a different meaning in the proteomic and composite stages and are listed once with both readings; the reuse is noted in §3.1 and does not affect any result.

Table 2: Notation used in this thesis.

Symbol	Meaning	Defined
g, c	Index over genes and over cell lines respectively	§3.3
$\hat{\sigma}_g$	Robust scale estimate for gene g (rescaled median absolute deviation)	Eq. (3.1)
L	Lineage stratum: the tissue of origin within which a gene’s scores are standardised	§3.3.1
n_L	Number of cell lines in lineage L with a finite value for the gene	Eq. (A.1)
$\tilde{\sigma}_{L,g}$	Lineage dispersion after shrinkage toward the gene-global estimate	Eq. (A.1)
w_L	Shrinkage weight $n_L/(n_L + n_0)$	Eq. (A.1)
n_0	Shrinkage constant in the lineage-dispersion weight w_L	Eq. (A.1)
$z_{g,c}$	Standardised evidence score for a gene and cell line	§3.3.1
$d_{L,g}$	Detection fraction: proportion of finite values at or above the floor, within lineage L	§3.3.1
q_{floor}	Quantile defining the per-method detection floor	§A.2.3
$q_{\text{upper}}, q_{\text{lower}}$	Detection-regime cut-points on $d_{L,g}$	§3.3.1
Z	Weighted inverse-normal combination of method- and source-level scores	§A.2
k_{src}	Number of sources contributing a finite score	§A.2.6
ρ_g	(i) Per-gene rank correlation between the two proteomic platforms; (ii) shrunk per-gene RNA–protein correlation	Eq. (3.2); Eq. (A.2)
λ	(i) Lower agreement bound mapping ρ_g onto κ_g ; (ii) shrinkage weight $K/(n_g + K)$	§3.3.2; Eq. (A.2)
v	Upper agreement bound mapping ρ_g onto κ_g	§3.3.2
κ_g	Cross-platform evidence-agreement factor for the protein arm; not a probability that a recommendation is correct	Eq. (3.2)
p_0, k	Hill half-maximum and steepness for a genomic evidence channel	Eq. (3.3)
p_{base}	Noisy-OR combination of genomic channels; a bounded ordering, not a calibrated probability	Eq. (3.3)

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Table 2 continued from previous page

Symbol	Meaning	Defined
b	Additive increment where a fusion preserves the reading frame	§A.2
$e_{g,c}$	Protein residual after removing the RNA-explained component	Eq. (3.4)
$e_{g,c}^*$	Residual standardised, shrunk and clipped for entry into the composite	§3.3.4
β_g	Ordinary least-squares slope of protein on RNA for gene g	Eq. (A.2)
ρ_0, K	Panel-wide correlation prior and its shrinkage constant	Eq. (A.2)
$N_{\min}$	Minimum shared cell lines before ρ_g is estimated for a gene	Eq. (A.2)
$W_{\text{RNA}}, W_{\text{prot}}$	Arm weights, each $\sqrt{n_{\text{eff}}}$	§3.3.4
n_{eff}	Kish effective sample size for an arm	§A.2
Φ	Standard normal cumulative distribution function	Eq. (3.5)
$s_{g,c}$	Composite score, bounded in $[0, 1]$	Eq. (3.5)
$\pi_{g,c}$	Within-gene percentile rank of the composite score, used for tier assignment	§3.4
τ	Percentile threshold for leading status and tier directional consistency	§A.2
J_c	Joint co-selection score	Eq. (3.6)
ϕ	Co-selection retention floor	Eq. (3.6)
Sel_c	Selectivity score for an exclusion query	Eq. (3.7)

Acronyms

Table 3: Acronyms used in this thesis.

Acronym	Meaning
RNA	Ribonucleic acid
RNAi	RNA interference
CRISPR	Clustered regularly interspaced short palindromic repeats
TPM	Transcripts per million
nTPM	Normalised transcripts per million
TMT	Tandem mass tag
DIA	Data-independent acquisition
OLS	Ordinary least squares
MAD	Median absolute deviation
CDF	Cumulative distribution function
Kish ESS	Kish effective sample size
AUROC	Area under the receiver operating characteristic curve
pAUC	Partial area under the curve
FPR	False positive rate
CVCL	Cellosaurus accession prefix
RRID	Research Resource Identifier
ENSG	Ensembl gene accession prefix
VEP	Variant Effect Predictor
CADD	Combined Annotation Dependent Depletion
FFPM	Fusion fragments per million
GEO	Gene Expression Omnibus
GSE, GSM	GEO series and sample accession prefixes
HPA	Human Protein Atlas
CCLE	Cancer Cell Line Encyclopedia
GDSC	Genomics of Drug Sensitivity in Cancer
ICLAC	International Cell Line Authentication Committee
API	Application programming interface
IAM	Identity and Access Management
JSON	JavaScript Object Notation

Abstract

AstraZeneca requested a method to identify cell lines that express a target gene, utilizing public molecular data to provide ranked, confidence-scored, and justified recommendations. The Cell Line Selector was developed to harmonize nineteen public datasets from seven data programmes, resulting in a resource comprising 2,127 cell lines and 20,163 protein-coding genes. Cell-line identity is resolved through stable accession numbers rather than names. Gene-expression and protein measurements are standardised against lines of the same lineage and integrated into a score reflecting target-gene expression strength, while genomic alterations determine a separate confidence tier. Missing measurements are not interpreted as low values, and molecularly similar alternative lines are screened against a register of problematic cell lines.

Testing with drug-response data showed that cell lines the model found to be similar reacted to drugs more alike than cell lines that only came from the same tissue. A subsequent re-audit retracted an earlier finding that low expression predicts the absence of CRISPR dependency, as shuffled labels reproduced 98% of the observed effect. The interface supports single-target, co-selection, and exclusion queries, while a constrained language-model agent explains scores and retrieves provenance without altering rankings. When evaluated against independent CRISPR and drug-sensitivity screens, the score excluded non-dependent lines above chance but ranked the remaining candidates at or near chance, indicating its utility as a filter rather than a fine-grained ranker. In the absence of an accepted ground truth for cell-line selection, every ranking is traceable to its supporting evidence but is not established as biologically correct.

Achievements

- Calibrated four scoring parameters through an exhaustive search over 480 settings, and derived proteomic agreement thresholds from null correlation distributions rather than setting them by hand.
- Built a Monte Carlo uncertainty layer reporting 95% intervals, top- k stability and win-rates for every candidate.
- Designed a 61-test suite across three tiers, including tests run against the deployed agent.
- Conducted a ten-part statistical re-audit combining shuffled-label controls, clustering-aware inference and RNA replication.
- Deployed the full system end to end: a React interface, a FastAPI ranking service, and a serverless agent on AWS Bedrock.

Chapter 1

Introduction

1.1 Motivation

Molecular oncology experiments start by selecting a cell line that faithfully reflects target biology. Three costly, recurrent failure modes have been identified.

If the target is absent, the experiment cannot address its research question. Knockdown in a cell line with low transcript abundance produces a null result, and transcript presence does not necessarily indicate protein expression. An inappropriate biological context may allow gene expression without capturing the relevant mechanism; in fusion biology, the fusion event itself is required rather than mere co-expression. Misidentification of cell lines renders results uninterpretable. Although prevalence estimates vary by sampling frame, all indicate a significant risk: the ICLAC Register lists 560 cell lines for which no authentic stock is known [2]; short tandem repeat profiling of a 278-line panel across 28 institutes found that 46% were cross-contaminated or misidentified [3]; and a recent review estimates that approximately one in five cell lines are affected overall [1].

Transcriptomic, proteomic, genomic, and metabolomic data relevant to these challenges are publicly available for numerous cell lines [8, 11]. The main challenge is integrating these diverse data types into robust, defensible selection criteria. Laboratories frequently rely on familiar cell lines, manual verification, or ad hoc scripts, approaches that do not produce ranked short-lists, supporting evidence, confidence metrics, or alternative options if the primary selection is unsuitable. Data sources often exhibit unstable identifiers, inconsistent measurement scales, and missing values. Additionally, data releases and cell line coverage evolve over time, resulting in naive aggregation that disproportionately favours frequently studied lines (Chapter 2). This study adheres to two guiding principles: absence of evidence should not be interpreted as evidence of absence, and the strength of evidence should be reported separately from its quantity.

1.2 Aims and Objectives

This research responded to an AstraZeneca brief to unify public transcriptomic and proteomic data (DepMap, Human Protein Atlas, GEO) for identifying cell lines with target gene expression, while excluding confounders. The brief required addressing data sparsity, heterogeneity, interpretable confidence scores, cell line similarity, and transparent justification. Two frameworks were evaluated: a classical statistical model for aggregation and confidence scoring, and an agentic model for structured justification.

The primary aim is to build and assess an auditable multi-omic framework for target-specific cell line prioritisation, defined by eight objectives detailed in Section 5.2: O1: Integrate diverse public multi-omic sources with provenance. O2: Independently standardise and model each

evidence layer. O3: Provide ranked recommendations with separate confidence reporting. O4: Support target, co-selection, and exclusion queries. O5: Quantify uncertainty. O6: Offer interpretation and provenance without changing rankings. O7: Suggest molecularly similar alternatives. O8: Assess biological usefulness. O8 remains only partly addressed due to the lack of a ground truth for cell line selection, not limitations of this work. Chapter 5 discusses this limitation and the remaining work for O8.

This work diverges from the original brief in three ways. First, it uses a relational store keyed on stable accessions (§3.2) instead of a knowledge graph, as the main challenges stem from measurements, not relationships. Second, while the brief envisioned an agent to prioritise cell lines, here the agent provides only interpretation and provenance for a fixed ranking (§3.6), with evaluation in §4.3. Third, certain modalities and features (e.g., single-cell transcriptomics, CRISPR screens, pathway activity, receptor status, assay compatibility) are excluded from the scored scope (§2.3.1); CRISPR data are used only as external evaluation labels (§4.2.3), not as ranking inputs.

1.3 Contributions

This research introduces a data-conversion system and rigorously evaluates its capabilities and limitations. First, it provides an auditable multi-omic corpus: nineteen public tables from seven programmes, each with distinct identifiers and measurement scales, are reconciled into 2,127 cell lines and 20,163 protein-coding genes across nine measurement types. Identity is resolved through stable accessions, and every data join is audited. Second, the system offers an evidence model aligned with measurement properties; each data layer is standardised according to its specific properties rather than being forced onto a common scale. Missing measurements add no evidence to the score, bound its attainable extreme, and are reported as reduced coverage. Third, the system enables ranking with separable score and confidence metrics, reporting relative abundance and genomic corroboration independently. Fourth, it delivers query-specific ranking with structural safeguards: target, co-selection, and exclusion queries employ tailored rules, and a structural safeguard prevents Y-linked genes in female and unknown-sex lines from being misinterpreted as evidence of absence. Fifth, the system implements an interpretation layer that cannot alter results; a bounded, tool-mediated agent provides explanation and provenance after ranking, and the architecture is evaluated at this boundary. Finally, the evaluation process refined the study’s primary claim: a ten-part re-audit revealed that an initially positive abundance-dependency result was largely reproducible under shuffled labels, and the statistical significance was overstated by approximately five orders of magnitude. The remaining claim is more narrowly and cautiously framed: a post-hoc subgroup effect that did not replicate in an independent RNAi dataset and is not established as biological (§4.2.3). The evaluation’s capacity to limit its own overstatements constitutes a contribution in itself.

1.4 Thesis Structure

Chapter 2 outlines the problem, describes each evidence layer, challenges of combining data sources, and existing systems. Chapter 3 presents the framework: harmonisation, layer-specific evidence modelling, composite score, confidence tiers, query semantics, and the explanation layer. Chapter 4 assesses each layer by its research question, reporting both negative and positive findings. Chapter 5 states the main claims, unachieved objectives, reasons, and future directions. Appendix A documents parameters, resolution rules, test coverage, and diagnostics for Chapters 3 and 4.

Chapter 2

Background

2.1 Chapter Overview

Cell-line selection is a critical early decision in laboratory experiments, yet current tools offer limited support. This chapter defines the problem in computational terms (§2.2), outlines molecular evidence layers (§2.3) and their public sources (§2.4), explains why direct integration is not feasible (§2.5), reviews methodological solutions (§2.6), and compares Cell Line Selector to existing systems (§2.7).

2.2 The Cell Line Selection Problem

2.2.1 Cell Lines as Experimental Models

A cell line is a population of cells from a single source, usually a tumour biopsy, adapted to grow indefinitely in culture. Their ease of use has made cell lines the primary experimental system in molecular oncology and the basis for large-scale screening [8, 11]. They were central to developing EGFR, BRAF, and ALK inhibitors [12]. This has created a vast public data landscape, with transcriptomic, proteomic, genomic, and metabolomic measurements for the same lines from different labs and technologies.

2.2.2 Consequences of Poor Model Selection

A cell line’s usefulness depends on how well it models relevant biology. Three main failure modes recur.

- **Target absent.** Knockdown in a cell line with low transcript abundance yields a null result, providing no information on gene function. Detecting transcript does not guarantee protein presence (§2.3).
- **Biological context incorrect.** A gene may be expressed in a context that does not reproduce relevant biology: a loss-of-function mutation may disable the pathway, and fusion-driven cancer research requires a cell line with the actual fusion event, not just partner gene expression.
- **Identity incorrect.** The ICLAC Register lists 560 cell lines with no authentic stock [2], about one in five in use is compromised [1], and a survey of 278 tumour cell lines found 46% cross-contaminated or misidentified [3, 4]. Reporting Research Resource Identifiers reduces problematic line use [5]; unverified reagents carry a high reproducibility cost [6].

Cell-line names are unreliable, but authentication is efficient and public registers exist [1, 7]. A ranking system must resolve identity using stable accession numbers, and check candidates

against public registers of problematic lines before offering them as substitutes for one another (§3.5.1).

2.2.3 Current Practice and Computational Reformulation

Currently, labs reuse existing cell lines (concentrating research on a few models), manually inspect databases (which does not scale or document rejections), or write one-off scripts (rarely reusable and limited to one data type). None produces a ranked shortlist, supporting evidence, confidence measure, or alternatives if the first choice fails.

Computationally, the task is ranked retrieval over heterogeneous, incomplete evidence. Queries specify genes to include or exclude; the corpus is the set of characterised cell lines; evidence consists of per-gene, per-line measurements from assays differing in scale, coverage, and bias. Information retrieval methods apply (§2.6), adapted to distinct sources that relate differently to biology.

2.3 Layers of Molecular Evidence

Transcription and translation are regulated independently, so measurements at one level do not predict outcomes at the next; each molecular layer offers distinct, complementary evidence. Table 2.1 summarises each layer’s question and limitation; the following paragraphs give the implications for ranking system design.

Transcriptomics. RNA sequencing quantifies gene transcripts, typically as transcripts per million, but sources use incompatible conventions: DepMap provides $\log_2(\text{TPM}+1)$ values from RSEM quantification [13, 17], while the Human Protein Atlas uses trimmed mean of M-values (nTPM) [14, 15], so a value of 5 does not mean the same thing in each. Transcriptomic coverage is comprehensive, making it the foundation for ranking systems. It cannot, however, confirm functional protein presence: transcript and protein levels correlate poorly [25, 26], and most residual disagreement reflects experimental reproducibility, not biology [27].

Proteomics. Two main datasets exist (CCLE Gygi and ProCan) and are not interchangeable: Nusinow et al. [19] profiled 375 lines via tandem mass tag labelling [24], and Gonçalves et al. [21] profiled 949 lines using data-independent acquisition [23]. Proteins found in the second dataset have higher average RNA expression, so absence from the matrix indicates weak evidence for low abundance rather than true absence, and imputing missing values as zero penalises low-level targets [21].

Somatic mutations and gene fusions. A single gene may have multiple mutation records in one cell line, so aggregating by gene is inappropriate, and different variants have distinct functional effects requiring detailed annotation. The COSMIC Cancer Gene Census provides evidence tiers [28, 29] that should be treated as separate categories. Gene fusions combine segments from two genes, potentially producing novel proteins; fusion calls are filtered upstream, so downstream analyses inherit prior decisions [17]. In fusion-driven analyses, presence of a fusion is both a filtering criterion and a scored feature.

Copy number, microRNA and metabolomics. Copy-number data quantify genomic region copies relative to a diploid baseline [32]. Amplification drives oncogene overexpression and homozygous deletion confirms gene absence, but measurements span segments, so resolution is coarser than gene-level. MicroRNAs regulate gene expression post-transcriptionally and partly explain transcript–protein disagreement [9]; the field lacks a standard analysis pipeline [33], so normalisation must be explicitly justified. H. Li et al. [34] measured 225 metabolites in

928 lines, but the matrix mixes measured and imputed values after batch correction, and gene expression is an unreliable predictor of steady-state metabolite abundance [35]; metabolomics therefore serves better as context than as supporting gene-level evidence. The consequences for microRNA and metabolomics are detailed in §2.3.1.

Derived features. Model-level features, such as aneuploidy and whole-genome doubling, can differ between quarterly data releases [18]. Recording the exact data release is essential for reproducibility.

Table 2.1: Conceptual evidence layers, the question each addresses, and the main limitation.

Layer	Answers	Principal limitation
Transcriptomics	Is the gene transcribed, and how strongly?	Weak predictor of protein presence
Proteomics (TMT)	Is protein present, relative to a reference?	Relative, not absolute; 375 lines
Proteomics (DIA)	Is protein present, and in what quantity?	Detection biased towards abundance
Somatic mutation	Is the gene altered, and how?	Consequence requires annotation
Gene fusion	Does a fusion involving the gene exist?	Inherits upstream filters
Copy number	Is the gene amplified or deleted?	Segment-, not gene-, resolution
miRNA	Is post-transcriptional regulation active?	No standardised pipeline
Metabolomics	What is the metabolic state of the line?	Poorly related to expression

No single layer determines whether a cell line is a suitable model for a gene: transcriptomics may rank a line that transcribes but does not translate the target, proteomics would exclude most lines for want of coverage, and mutation data say nothing about abundance. Integration is required because each layer answers a different question.

2.3.1 Scope of the Scored Evidence

Not all the layers described are ranked. Transcriptomic, proteomic and genomic evidence are standardised and scored (§3.3). Copy number is used as a directional modifier for confidence, not for scoring (§3.3.3). MicroRNA and metabolomic data are harmonised and included in coverage (Table 3.1) but are *not* scored or evaluated here: neither has a justified normalisation convention, and the metabolite matrix mixes measured and imputed values [34], conflicting with the principle that inferred values should not be presented as measurements. Keeping these data unscored preserves provenance completeness and avoids unsupported claims.

2.4 Public Data Resources

The corpus uses seven public programmes, differing in measurement focus, permitted use, and integration. Two considerations influence system design.

First, licensing restrictions determine permissible redistribution. For example, COSMIC allows academic use with registration but prohibits redistribution, so products must exclude COSMIC-derived fields or restrict distribution. This constraint determines which evidence can be included with recommendations.

Second, some resources have addressed parts of integration. Cell Model Passports was developed for the identity problem (§2.2.2); its authors note neighbouring resources are insufficient:

Cellosaurus and cBioPortal lack functional data, and PharmacoDB lacks genomic annotation [12]. The Human Protein Atlas checks each cell line for consistency with its TCGA tumour cohort to support model selection [16]. This prior art, discussed in §2.7, supports treating identifier resolution as a core system component, not just preprocessing (§3.2).

Among the seven sources, transcriptomic datasets are the most heterogeneous because GEO aggregates independent depositions, each with its own platform, summarisation, and scale. §3.3.1 standardises within method, not source. The GEO component of this project is a multi-series Affymetrix compendium, with the main normalisation families being RMA/gcRMA (log-scale) and MAS5 (linear), classified per sample/series rather than applied uniformly across all GEO data.

2.5 Challenges in Cross-Source Integration

Although these datasets describe the same cell lines, direct integration is not feasible. Five obstacles hinder integration, each with direct implications for system design.

- **Entity identity.** Cell-line names lack uniqueness and stability; resources use different, sometimes inaccurate, identifiers, which change over time; for example, recent DepMap releases record old identifiers when models are merged [17]. Ignoring aliases leads to silent data loss. Identifier resolution must be a core architectural component, not a preprocessing step.
- **Incommensurable scales.** Data layers use different measurement types: log-transformed expression, label-free protein intensities, imputed metabolite abundances, integer copy numbers, and categorical variants. Arithmetic across them is not meaningful; evidence must be converted to a common representation before integration.
- **Structured missingness.** Missing values have three causes, each needing different treatment. Proteins may be missing because abundance is below detection (not-at-random) [21, 39]; a measurement may be missing because the cell line was not studied, or the value was imputed. Structural absence should add no evidence to the score and be reported as reduced coverage; treating all missingness as zero is a major modelling error.
- **Batch effects and version drift.** Source studies correct for batch effects [40], but version drift is less discussed: quarterly releases change values, and derived features differ across quarters [18]. Recording provenance and fixing data versions are essential for reproducibility.
- **Coverage asymmetry.** Sources characterise very different numbers of cell lines (Table 3.1). Left unaddressed, a difference in how much evidence exists for a candidate is indistinguishable from a difference in how strong that evidence is. Coverage must therefore be recorded and reported, not absorbed into the score.

2.6 Methodological Foundations

Each obstacle above has an established methodological response.

- **Within-source normalisation.** Each source is pre-normalised by its own conventions [13, 14, 20, 22]. Recognising prior transformations avoids errors like reapplying logs to already log-transformed data.
- **Relative scoring.** Layers are compared using measurements relative to a reference population, not native units. Percentile ranking is robust but loses magnitude and can rank non-expressed genes, so an absolute threshold is also needed. The system uses robust standardisation with a detection floor for evidence layers (§3.3.1) and reserves percentile rank for tier assignment (§3.4), comparing only available lines for each gene.

-
- **Directional alignment.** Whether evidence is supportive or contradictory depends on the query: high target expression supports selection, high excluded-gene expression does not. Directionality is set at query time, shaping the architecture.
 - **Ranking.** Weighted linear combinations show each layer’s contribution, but arbitrary weights lack justification. Training-free rank fusion handles missing data [42], but we did not use it because it discards magnitude and coverage needed for confidence, and was not evaluated here. Learned combination is not possible without an accepted ground truth for cell-line selection (see Chapter 4).
 - **Confidence.** Confidence is not a single value, as different doubts require different responses. Two components are used: *coverage*, which records how much evidence exists for a candidate and is reported alongside the score rather than absorbed into it, and *concordance*, which records whether abundance and genomic evidence agree and is expressed as the confidence tier (§3.4). *Margin* is important but not reported; interval overlap and rank stability (§4.2.2) imply margin, but no statistic is given. This is a limitation.
 - **Similarity and alternatives.** Suggesting substitutes is a nearest-neighbour problem in high-dimensional space, where distances concentrate [45, 46]; restricting to biologically relevant genes helps. Similar profiles do not guarantee interchangeability: cross-contaminated lines look similar because they are identical [3]. Similarity functions must be identity-aware to avoid recommending contaminants as substitutes.

2.7 Related Work and Positioning

Source portals are authoritative for their own data but do not integrate across sources; combining DepMap expression with ProCan proteomics still requires manual reconciliation.

CellMinerCDB integrates pharmacological and genomic analyses across several cell-line panels [47, 48], and its authors note the same obstacle described here: naming and processing differences that portals do not resolve [48]. It supports exploration, not selection; users interpret the analysis, and it produces no ranked recommendations. PharmacDB aggregates pharmacogenomic studies and, by its authors’ account, has converged with CellMinerCDB and DepMap on drug-response exploration [49].

ShinyTHOR is closest in motivation, providing multi-omic access across CCLE and related resources for researchers without the expertise to extract the data themselves [50]. One module ranks the top and bottom ten cell lines by marker expression, using a single modality. Cell Line Selector differs in ranking across multiple evidence layers, reporting explicit confidence decomposition, and recording which evidence produced each position. It is not, however, the first to produce an ordered list for a stated marker.

Model-fidelity approaches ask a different question: which cell line most closely resembles a given tumour type [16, 52], operating at the transcriptome and cohort level. These complement rather than compete with this work, since gene-targeted and tumour-similarity rankings serve different purposes.

MOSA integrates DepMap multi-omic data and adds synthetic values, using SHAP for explainability [51]. Its purpose is imputation and discovery, whereas this work concerns selection; for selection, generating values never measured is a liability that may mislead researchers. Latent-variable models likewise trade transparency for expressiveness, making attribution harder for bench scientists than direct statements of evidence contributions.

2.7.1 Outstanding Challenges

Five activities stand out: data access (source portals), data exploration (CellMinerCDB and PharmacODB), multi-omic integration (ShinyTHOR and MOSA), tumour-model similarity (Human Protein Atlas and related work), and target-specific ranked cell-line selection.

Within the literature and systems reviewed for this thesis, no identified framework combined quantitative molecular ranking with explicit evidence attribution, missing-data handling and provenance-aware downstream explanation in the form implemented here.

2.8 Chapter Summary

Cell-line selection determines both the relevance and reproducibility of experimental results, yet it often relies on convention or manual inspection; the prevalence of misidentification shows that unverified selection carries measurable risk. The relevant evidence spans multiple molecular layers, each answering a distinct question, so integration is necessary, but the five obstacles in §2.5 show it cannot be achieved arithmetically, and the responses in §2.6 impose constraints on the design that follows.

Three of these constraints shape the whole of Chapter 3. First, identity must be resolved through accessions before any integration occurs. Second, evidence must be standardised within each molecular layer before being combined across layers. Third, missing measurements should add no evidence to the score and be reported as reduced coverage.

Chapter 3

Execution

3.1 Overall System Architecture

Given a molecular target, which cell lines serve as the most suitable experimental models? Answering this requires integrating evidence that does not naturally align: nineteen public datasets from seven programmes describe overlapping sets of cell lines using different identifiers, row granularities and measurement scales, and were not designed for integration.

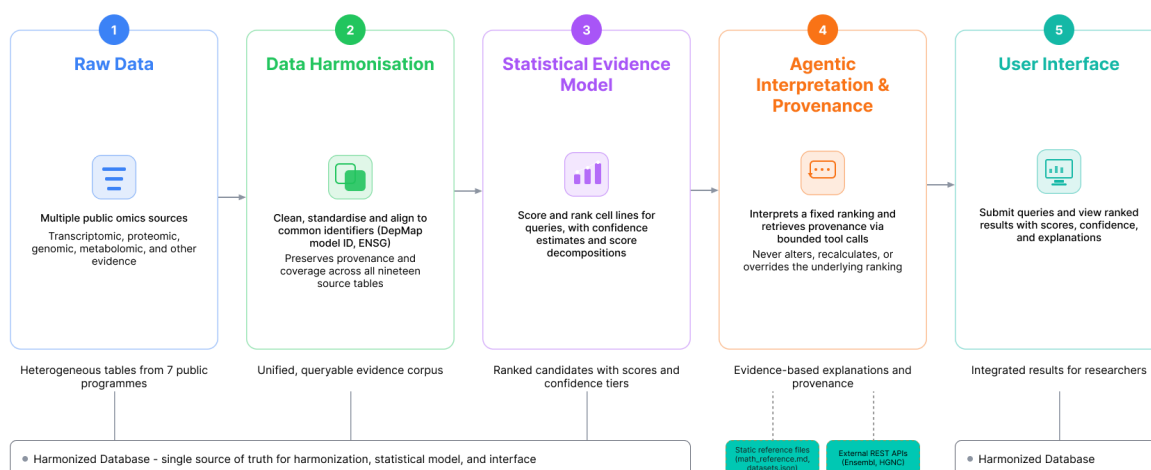


Figure 3.1: System architecture. Panel 1 is the raw input; panels 2–5 are the four processing stages.

The four processing stages in Figure 3.1 are data harmonisation, which standardises all sources into a common representation; statistical evidence modelling, which converts each measurement type into a comparable quantity, integrates continuous evidence into a score, and classifies each candidate by degree of genomic corroboration; agentic interpretation, which supplies explanation and provenance; and the user interface, which displays the ranking and its supporting evidence.

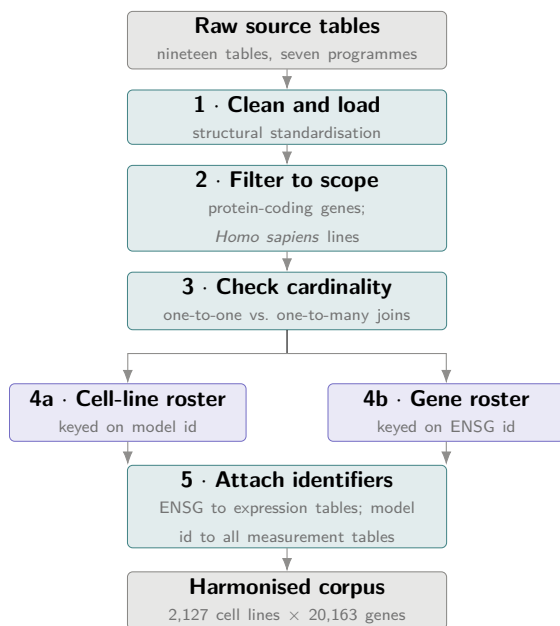
The statistical pipeline generates the ranking, and the agent does not modify, recalculate or override it. The harmonised database at the base of Figure 3.1 is the exclusive interface between raw evidence and all downstream processes: neither the statistical model nor the agent accesses source tables directly.

Two terms are used throughout and should not be conflated. A *score* quantifies the degree to which a cell line matches the specified target, on continuous transcriptomic and proteomic evidence. The confidence tier indicates whether genomic evidence corroborates that score; evidence availability is reported alongside it as coverage. A high score resting on limited evidence and a high score corroborated by robust evidence are reported differently.

Table 2 summarises the symbols used below. Two are context-dependent: ρ_g is the per-gene rank correlation between the two proteomic platforms in Eq. (3.2) and the shrunk per-gene RNA–protein correlation in Eq. (A.2); λ is an agreement bound in §3.3.2 and the empirical-Bayes shrinkage weight $K/(n_g + K)$ in Eq. (A.2). This reuse does not affect the mathematical results and is noted here to prevent confusion.

3.2 Data Harmonisation

The first stage established a common representation before any evidence was scored. Figure 3.2 sets out that sequence.



Four kinds of heterogeneity required resolution. Shape: measurements were stored one per row, one per column, or transposed with genes as rows. Granularity: some tables were keyed by sequencing profile rather than by cell line, so one model appeared in several rows. Identifier format: no universal key existed across tables, and the same accession appeared in incompatible forms CVCL_2063, cvcl:2063, RRID:CVCL_2063 all denoting one cell line. Measurement scale: log-transformed expression, relative protein and metabolite abundances, integer copy number, and categorical variant classes. The first three are addressed here; the fourth in §3.3.

Figure 3.2: Data harmonisation workflow, from the nineteen raw source tables to the harmonised corpus. Resolution rules, cardinality counts and validation outcomes are given in Appendix A.1.

3.2.1 Identity, Granularity and Scope

DepMap model accession was chosen as the standard cell-line identifier because it refers to the biological model rather than to sequencing runs, appears in six of nineteen tables, and is supported by DepMap’s alias and merge records [17]. Resolution is hierarchical: DepMap accession, then Cellosaurus accession [7], then free-text name if needed. Controlled identifiers are prioritised because names are ambiguous an audit found 580 names shared by human and non-human Cellosaurus entries, so filtering by name could exclude valid records. Human cell lines are therefore restricted by accession, not by name. Granularity was treated as a schema decision rather than a data-cleaning rule. Two authentic one-to-many relationships exist: a model corresponds to multiple sequencing profiles, and a cell line to multiple GEO samples across studies. Alternative identifiers are stored as sets while model accessions remain unique within the roster, so every table-to-roster join is one-to-one. Where a single identifier matches several models, the record is duplicated and each instance flagged, allowing those rows to be excluded during sensitivity analyses rather than dropped or arbitrarily assigned. Appendix A.1 gives resolution rules, cardinality counts and ambiguity volumes.

Two further decisions preserved information for analysis. Missing values were not imputed, keeping the distinction between attempted and unattempted measurements. Outliers were kept, as extreme values may be meaningful. Both principles apply throughout the evidence model.

3.2.2 The Harmonised Corpus

The corpus comprises 2,127 cell lines and 20,163 protein-coding genes measured across nine distinct types (Table 3.1). Coverage varies substantially, from 91.9% for genomic signatures to 17.6% for proteomics. The proteomics (Gygi) row counts the tandem mass tag platform only, at 375 cell lines; the data-independent acquisition platform of §3.3.2 (PROCAN) covers a larger panel [21], and the two are merged symmetrically downstream. In total, 237 cell lines carry all nine measurement types and 61 carry none the latter enter the roster through identifier resolution but receive no score in any evidence layer, while remaining in the denominator of every coverage figure in Table 3.1. This heterogeneity is methodologically significant; §3.4 details how it is prevented from biasing the ranking.

The three transcriptomic sources are retained separately rather than merged. The 525 cell lines (24.7%) shared between the DepMap and GEO expression tables provide independent measurements of the same quantity, enabling assessment of cross-source agreement.

Table 3.1: Coverage of the harmonised corpus across the nine measurement types.

Measurement type	Lines	Cov.
Genomic signatures	1,955	91.9%
Somatic mutations	1,744	82.0%
Gene fusions	1,698	79.8%
DepMap expression	1,479	69.5%
HPA expression	1,105	52.0%
Metabolomics	927	43.6%
microRNA	914	43.0%
GEO expression	590	27.7%
Proteomics (Gygi)	375	17.6%
Proteomics (PROCAN)	949	44.6%

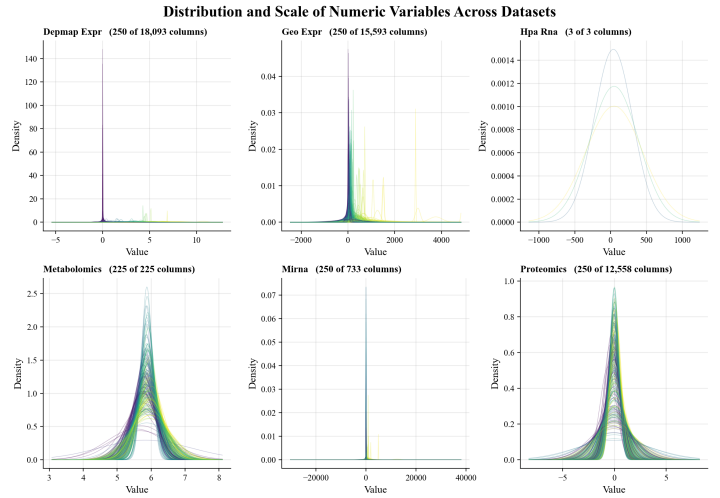


Figure 3.3: Density distributions for representative numeric variables from six harmonised measurement sources.

Figure 3.3 shows why these scales cannot be combined directly. The six panels share no common axis: DepMap expression, GEO expression, microRNA and metabolite values span ranges differing by several orders of magnitude, two panels show a pronounced spike at zero, and two are heavy-tailed. Averaging, summing or thresholding across these raw values therefore has no consistent interpretation the incommensurable-scales obstacle of §2.5. Harmonisation fixes the provenance and category of each measurement and makes it traceable to source, but does not make measurements from different platforms statistically comparable; that is the function of the evidence model.

3.3 Statistical Model

A log-transformed transcript abundance, a relative protein quantity and a categorical variant class cannot be pooled directly. Each measurement is first converted into a value expressing how unusual it is for the cell line in question, and only then combined. Four principles govern this.

1. Standardise within a layer before combining across layers: Raw measurements are not incorporated directly into the composite score.
2. Score and confidence are separate outputs: Evidence strength and evidence availability address distinct aspects of the evaluation.

3. Absence is not low abundance: an unmeasured layer adds no evidence in either direction. It enters the composite at the neutral value of its standardised scale, which bounds the attainable score without pushing it up or down (§3.3.4), and is reported as reduced coverage. Zero-filling raw abundance would instead introduce artificial negative evidence, particularly for low-expression targets, where precise selection is most critical.
4. Every score decomposes: Each ranking position must be attributable to specific, inspectable contributions, enabling recommendations to be justified from retrieved evidence rather than generated data.

3.3.1 Transcriptomic Evidence

Transcriptomic evidence comes from HPA, DepMap and GEO. After harmonisation these are addressable but not comparable, because their normalisation histories differ; GEO is especially heterogeneous, aggregating multiple platforms and algorithms onto unrelated scales, so direct pooling would mix technical with biological effects. Four parameters (q_{floor} , n_0 , q_{upper} , q_{lower}) are tuned by cross-source concordance (Appendix A.2).

Method resolution and scale harmonisation. Standardisation is performed by method defined by platform and processing family not by source. HPA and DepMap each provide one method; GEO provides one per unique combination, giving seven GEO methods: the six of Table A.3, one of which is the residual unknown family, together with the method reserved for samples carrying no metadata record. Processing family is extracted from metadata by lexical rules; each method is then profiled from its own data, with metadata acting as a prior and the data prevailing where the two disagree. All methods are rescaled to a common log scale, native units differing by over an order of magnitude. A detection floor is set per method: one TPM for RNA-seq sources, matching HPA, and a calibrated quantile otherwise.

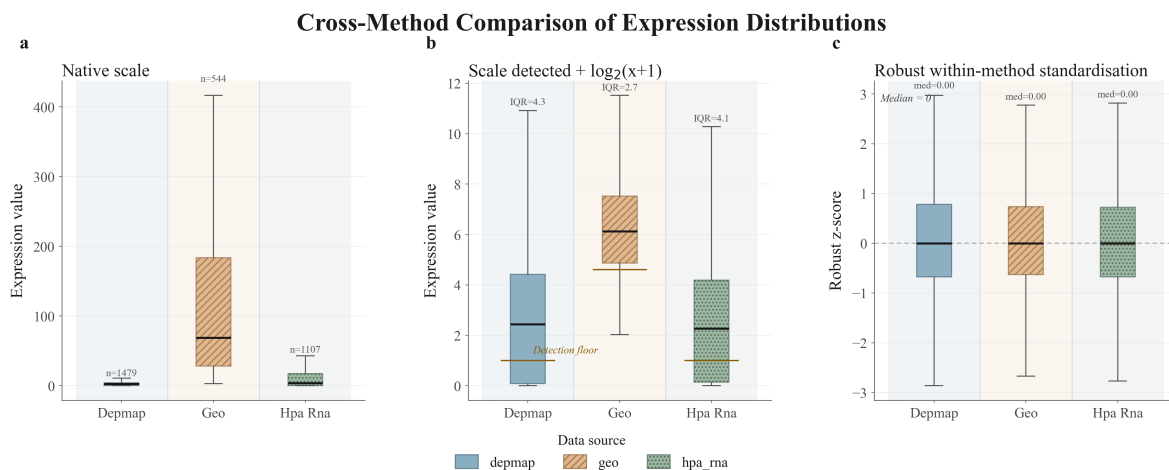


Figure 3.4: Expression distributions across the three transcriptomic sources: (a) native scale, (b) after empirical scale detection and $\log_2(x + 1)$ transformation, with the detection floor marked, and (c) after robust within-method standardisation.

Figure 3.4 shows the effect of this sequence. In native units the three sources are not comparable; after scale detection and $\log_2(x + 1)$ transformation they occupy a common range with similar interquartile widths and a visible detection floor; after robust within-method standardisation all three are centred at a median of zero. The first panel establishes that harmonisation is a precondition for meaningful comparison, not a refinement of it.

Relative scoring. Establishing a common scale is not itself evidence. What matters is whether a line expresses a gene *unusually strongly relative to comparable alternatives*. The

classical z -score captures this, but neither its standard estimators nor its default reference population is appropriate here.

Expression distributions contain amplified and silenced outliers that the platform is designed to detect, and a single overexpressing line can inflate the standard deviation and so compress all other scores. The median and median absolute deviation (MAD) are used instead, both resistant to contamination affecting up to half the reference population, with the MAD rescaled by the standard consistency factor to retain the familiar z -score interpretation [61]:

$$\hat{\sigma}_g = 1.4826 \cdot \text{med}_i |x_{g,i} - \text{med}_j x_{g,j}|. \quad (3.1)$$

Equation (3.1) gives the scale estimator only; the score is the deviation of a measurement from its stratum centre divided by this estimate. The constant 1.4826 makes the estimator consistent for the standard deviation under normality, so the familiar z -score interpretation holds to the extent that assumption does. The estimator is defined per gene but applied to (gene, lineage) strata, using lineage rather than the full corpus as the reference: because tissue of origin drives expression variation, a global score for lineage-restricted genes would mostly reflect tissue rather than within-lineage differences. Global scores are therefore secondary, used only for cross-lineage queries. Because stratum sizes vary, dispersion estimates from small strata are shrunk toward the global method estimate, and with fewer than five lines the centre defaults to the global median. Because a stratum in which most values tie at the detection floor can yield a zero median absolute deviation, the shrunk estimate is floored away from zero (Appendix A.2), so the divisor is guaranteed positive. Each score records which estimate it used, so any borrowed scale is identifiable in queries.

Detection regimes. Standardisation assumes continuous variation across the reference population. For genes detected in only a few lines within a lineage, most values sit at the detection floor and differ mainly by noise, which a parametric estimator could misread as signal. The estimator is therefore selected per gene and lineage by the proportion of values above the detection floor. Where detection is high, the robust z -score is used. For partially censored data, a rank-based inverse-normal transformation is used [60], assigning floor-tied values a shared midrank and common score, which represents a bound rather than a point estimate. Where detection is very low, no score is reported; these withheld cases reduce source count without implying typicality. These three detection regimes are distinct from the composite-stage variants in §3.3.4. Because the estimator is selected per gene and lineage, scores are calibrated within a gene rather than across genes; §3.5 states the consequence for multi-gene queries.

Combination. Method-level scores are combined by the weighted inverse-normal method [62, 63] in two stages: first within each source, weighted by replicate count; then across the three sources with equal weight. The separation is intentional. GEO contributes six methods against one each from HPA and DepMap, so weighting by method count would amplify the least robust evidence on account of its internal heterogeneity rather than its reliability.

3.3.2 Proteomic Evidence

Two independent programmes quantify protein abundance by distinct technologies [23, 24], measuring the same quantity on unrelated scales, with neither serving as a definitive reference. Where transcriptomics involves a single source diverging into multiple methods, proteomics must reconcile two sources without privileging either.

Gene-level resolution and isoform collapse. Both matrices are indexed by protein accession while the rest of the system operates at gene level, so accessions are first resolved to

the harmonised gene roster; 96.7–96.8% resolve successfully. Multiple accessions often map to a single gene, but cannot be assumed to be independent measurements: they may be isoforms of one protein detected with varying efficiency, or distinct products with unrelated abundances. These require opposite analytical treatments and cannot be distinguished from identifiers alone, so concordance is assessed empirically. Pairwise Spearman correlations are calculated across shared cell lines and the minimum correlation compared against a derived threshold, making concordance conjunctive, so that a single divergent accession prevents the gene being treated as one quantitative entity.

Where accessions are concordant, the gene takes the maximum value; this is justified only because the test has shown all accessions to measure a single quantity at varying detection efficiencies, making the largest value the least attenuated. Where accessions are discordant or unassessable, the mean is used, preventing any single accession dominating. A lone accession is used unchanged.

Independent standardisation. Each platform is standardised independently using the lineage-conditioned robust estimator of §3.3.1, because one platform reports abundance relative to an internal reference, giving values not comparable even across its own batches. At least five finite values per lineage are required before a score is generated, preventing under-observed columns from producing defined but meaningless centres and scales. Scale-estimation fallbacks and the derivation of the concordance threshold and agreement bounds λ and ν all estimated from null and matched correlation distributions are given in Appendix A.2. Figure 3.5 shows that attenuation narrows the protein evidence distribution without shifting its centre, and increases the exact-zero proportion from 36.0% to 40.8%.

Coverage and agreement play distinct roles. Once both platforms are scaled identically, two independent quantities govern their merging. *Coverage* sets influence: each platform is weighted by its number of cell lines with finite scores, gaining influence through broader measurement. *Agreement* sets confidence: the per-gene rank correlation ρ_g between platforms is mapped to an attenuation κ_g , interpolated between bounds λ and ν estimated from the data. Rank correlation is used because consistent ranking, not numerical agreement, is what matters. Equation (3.2) defines both: $z_{g,c}^{\text{raw}}$ is the coverage-weighted merge of the two platform scores, and the protein evidence carried forward is that merge attenuated by κ_g . The two remain conceptually distinct: coverage sets each platform’s weight in the merge, and agreement sets how much of the merged protein evidence reaches the composite score.

$$z_{g,c}^{\text{raw}} = \frac{\sum_s \sqrt{n_{g,s}} z_{g,c,s}}{\sqrt{\sum_s n_{g,s}}}, \quad \kappa_g = \text{clip}\left(\frac{\rho_g - \lambda}{\nu - \lambda}, 0, 1\right), \quad (3.2)$$

with $\kappa_g = 1$ where ρ_g cannot be estimated, so an absent agreement estimate is not treated as evidence of disagreement. κ_g is an evidence-agreement factor, not a probability that a recommendation is correct. When platforms disagree for a gene, the merged value is attenuated rather than resolved in favour of one source. Attenuation applies only where both platforms report the same gene and cell line; single-platform observations are not penalized for discrepancies elsewhere. If κ_g is clipped at zero, a gene with no platform agreement contributes zero protein evidence—the same as when no proteomic data exists (§3.3.4). These states are distinguishable in the stored output by platform count and pre-attenuation score (Appendix A.2.6), but not in the composite itself.

Thresholds are derived, not asserted. The concordance threshold, the agreement bounds λ and ν , and the minimum overlaps are all estimated from data. The concordance threshold

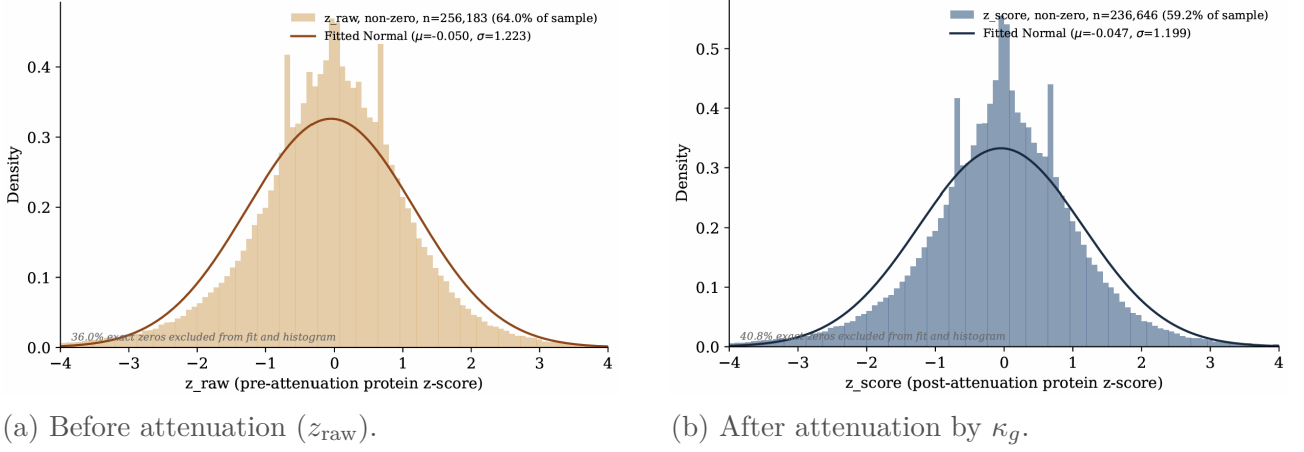


Figure 3.5: Distribution of non-zero protein z -scores (a) before and (b) after attenuation by the cross-platform agreement factor κ_g , each against a fitted normal reference.

and λ use null distributions of correlations between unrelated pairs, so the threshold reflects real agreement; the upper bound v uses matched same-gene correlations, so confidence saturates at typical matched-pair agreement; minimum overlaps are set for statistical power, excluding correlations with low sample size. Only the significance level and reading percentiles are fixed by convention.

3.3.3 Genomic Evidence

Genomic evidence differs from other types. Mutation consequences, pathogenicity predictions, variant counts, and structural calls lack a common unit, so joint standardisation is inappropriate. Each channel is mapped to $[0, 1]$ with a bounded transformation, and channels are combined by a noisy-OR: any strong channel raises the result, while unobserved channels leave it unchanged [57]:

$$h(x; p_0, k) = \frac{x^k}{x^k + p_0^k}, \quad p_{\text{base}} = 1 - \prod_i (1 - p_i). \quad (3.3)$$

Here p_0 is where a channel contributes half its maximum effect; unavailable channels are set to $p_i = 0$ and act as the identity. Because channels are not independent, the score may overstate evidence compared to a dependence-aware model. It provides a bounded *ordering* of evidence, not a calibrated probability. Equation (3.3) is applied to mutation and fusion channels, giving p_{mutation} and p_{fusion} (Figure 3.6). A gene–line pair is flagged `has_driver_alteration` if either value exceeds the threshold or a role-consistent copy-number event is present. Mutation channels are predicted consequence, predicted pathogenicity, and within-line recurrence. Fusion channels are caller confidence, expression support, reproducibility across samples, and a small increment for preserved reading frame [31, 53, 54, 55, 58]. Channels use raw values, not percentiles, as percentiles can inflate scores regardless of evidence strength.

Figure 3.6 shows evidence is sparse: variant evidence appears in 3.58% of rows, fusion evidence in 0.81%. When present, it is usually decisive: 90.7% of variant and 64.8% of fusion evidence exceeds the driver threshold. Genomic evidence corroborates only a minority of candidates and is assigned to the confidence tier, not the main score.

Copy number is handled separately, because its meaning depends on gene function: amplification matters for oncogenes but not for tumour suppressors, and homozygous deletion the reverse. A single value cannot capture this, so copy number is represented by role-gated flags based on Cancer Gene Census roles [28, 29]. Events matching a gene’s role are flagged as alterations; those in the opposite direction, or in genes of unknown role, are kept as context without corroborative weight. Copy number does not change the genomic evidence score but modifies the confidence tiers. Channel parameters and copy-number rules are given in Appendix A.2.

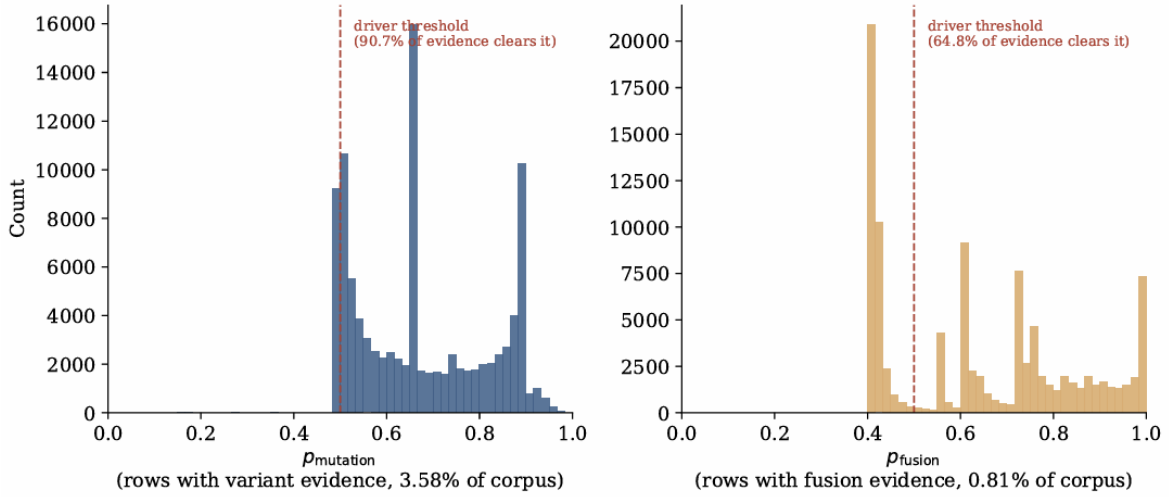


Figure 3.6: Distributions of the combined genomic evidence scores for mutations (p_{mutation} , left) and fusions (p_{fusion} , right), over the rows for which each kind of evidence exists.

3.3.4 Composite Score

The composite stage integrates continuous evidence while maintaining genomic evidence as a distinct interpretive axis.

The principal difficulty is the non-independence of RNA and protein measurements. Transcript abundance partially determines protein abundance, so averaging the two would double-count shared variance and diminish the unique contribution of the protein data. Protein is therefore residualised against RNA: for each gene, a regression of protein on RNA is fitted across cell lines with both measurements, and only the residual component of the protein evidence is used.

$$e_{g,c} = z_{g,c}^{\text{prot}} - \beta_g z_{g,c}^{\text{RNA}}. \quad (3.4)$$

Protein evidence thus contributes information beyond transcript abundance rather than a redundant measurement. The slope is the ordinary least-squares coefficient rather than the correlation, because dispersion differs substantially between the two data types (Eq. (A.2)), and per-gene correlations are shrunk toward a panel-wide prior by a fixed-weight linear rule of the James–Stein type [64], the shrinkage constant being set rather than estimated (Appendix A.2). Both data types enter the composite in stabilised form. The RNA score is winsorised at ± 5.0 before combination; the residual from Eq. (3.4) is standardised per gene, shrunk toward the panel-wide residual dispersion, and similarly clipped, giving $e_{g,c}^*$ as distinct from the raw residual.

Parameter values are in Appendix A.2. Unless a query is explicitly cross-lineage, $z_{g,c}^{\text{RNA}}$ denotes the lineage-conditioned score of §3.3.1; the global score is substituted only for cross-lineage queries, and the choice is recorded with each result. The RNA score and the standardised residual are then combined and mapped to a bounded scale:

$$s_{g,c} = \Phi \left(\frac{W_{\text{RNA}} z_{g,c}^{\text{RNA}} + W_{\text{prot}} e_{g,c}^*}{\sqrt{W_{\text{RNA}}^2 + W_{\text{prot}}^2}} \right). \quad (3.5)$$

Each data type is weighted by the square root of its effective independent observations (Appendix A.2), so influence reflects independent evidence, not source count. Three correlated transcriptomic and two proteomic platforms receive similar weights, with protein evidence—covering fewer lines receiving slightly more. The monotone transformation Φ preserves ordering and produces scores in $[0, 1]$ for exclusion queries (§3.5). Residualisation in Equation (3.4) makes the two terms approximately uncorrelated over the cell lines used to fit β_g , justifying root-sum-of-squares normalisation. Since standardised inputs are not exactly unit-variance (Appendix A.2),

mapped scores cluster near the upper bound for strongly supported candidates; §4.2.2 quantifies this. The interface reports rank and exact values, not relying on visual score spacing (§3.7).

Equation (3.5) is the default combination, but alternatives exist. When both data types signal strongly ($|z| > 1$) in opposite directions, residualisation is unsuitable, so the composite uses standardised evidences directly (see audit column, Appendix A.2). The chosen variant depends on the observation, and values near $|z| > 1$ may shift with small changes; §4.2.2 quantifies this. These variants are distinct from the detection regimes in §3.3.1.

Missing evidence is handled explicitly. If a cell line lacks proteomic data, the protein term takes the neutral value $e_{g,c}^* = 0$ rather than being omitted. This keeps single- and two-layer observations on the same scale; otherwise, identical RNA evidence would yield more extreme single-layer scores, potentially inverting rankings. The value used is the mean of the standardised residual, so it adds no evidence but bounds the outcome: the denominator is unchanged, so a line without proteomic measurement cannot reach the extremes available to lines with corroborating protein evidence. Given the coverage asymmetry in Table 3.1, this is a systematic property of the ranking, so coverage is reported alongside every score, not folded into it.

3.4 Confidence Layer

The composite score and genomic alteration flags answer different questions and are kept separate. The score measures relative abundance; the flags indicate if a gene is genomically altered. A high score without alteration and a moderate score with a driver alteration require different interpretation.

Candidates are classified on two axes (Table 3.2): score and alteration. The score axis uses percentile rank within each gene, comparing only lines available for the queried gene to avoid bias from gene abundance. A candidate is *leading* if its within-gene percentile reaches the tier threshold τ , and *directionally consistent* if it reaches τ for gain-of-function alterations or falls at or below $1 - \tau$ for loss-of-function alterations (Appendix A.2). The alteration axis is directional: gain-of-function is supported by high abundance, loss-of-function by reduced abundance. Treating high abundance as corroborative in both cases would misread contradictions as agreement.

Table 3.2: Confidence tier definitions and their interpretation.

Tier	Score	Alteration	Interpretation
HIGH	Directionally consistent	present	Abundance and genomic evidence agree
MEDIUM	Leading	absent	Strong abundance evidence, no genomic corroboration
CONTEXT	Directionally inconsistent	present	Alteration present, abundance not in the expected direction
LOW	Not leading	absent	Neither axis supports the candidate

The tiers are not an ordinal ranking but indicate how evidence supports or contextualises the abundance score. The **CONTEXT** tier does not imply weaker evidence; it flags candidates whose abundance defies expectation for review, not dismissal. Table 3.2 is a grid, with tiers reflecting evidence support, not score magnitude. Whether these tiers predict recommendation accuracy is untested (see §4.2.3).

3.5 Recommendation

The evidence model generates a score and a tier for each gene and cell line. Converting these into a recommendation depends on the research question, and three query types are supported.

Single target. Candidates are ranked directly by the composite score $s_{g,c}$, with the confidence tier reported alongside.

Co-selection. Where an experiment requires several genes to be present, a candidate is only as good as its weakest target:

$$J_c = \min_{g \in G} s_{g,c}, \quad \text{retained if } J_c \geq \phi, \quad \phi = 0.50. \quad (3.6)$$

The minimum in Equation (3.6) enforces conjunctive query semantics; a mean would let strong evidence for one gene offset weak evidence for another, returning candidates that meet the query on average but not individually. The threshold excludes candidates that meet the ranking but not the requirement, so if no candidate is adequately supported, the method returns no results. $\phi = 0.50$ is a chosen threshold to the midpoint of the bounded score is not calibrated as like the transcriptomic parameters of Appendix A.2.3. A candidate is retained only if a score exists for every requested gene; if any are missing under §3.3.1, the candidate is not scored (§4.2.1). Because $s_{g,c}$ is calibrated within a gene, ϕ expresses a position relative to lines measured for that gene, not absolute abundance. Both factors are within-gene quantities, so Sel_c ranks by relative rather than absolute absence of the excluded gene; where that gene is detected in too few lines for a score to be defined, none is emitted and the candidate is not promoted on that basis (§3.3.1).

Exclusion and selectivity. Where an experiment requires one gene present and another absent, candidates are ranked by

$$\text{Sel}_c = s_{a,c} \cdot (1 - s_{b,c}). \quad (3.7)$$

Because $s_{g,c}$ is bounded within $[0, 1]$, the complement gives a clear measure of the excluded gene's absence, and the product is maximised only when both conditions hold. A difference would instead reward a gap, possibly favouring moderate target expression with no excluded-gene expression over strong target expression with minimal excluded-gene expression, though the latter is the better system. A line from a female donor lacks a Y chromosome, so a Y-linked gene is structurally absent, not merely unexpressed. Without correction, this would suppress candidates in co-selection and wrongly promote them in exclusion queries, since Equation (3.7) rewards absence of the excluded gene. Scores for Y-linked genes are withheld in female and unknown-sex lines, following the principle that unverifiable values are withheld, not imputed.

3.5.1 Alternative Cell Lines

Three axes: RNA, metabolomics and miRNA are built independently. For each, features missing in over 20% of lines are dropped and remaining values are median-imputed; up to 2,000 of the most variable features are kept, data are standardised, and correlations between lines are calculated. Imputation is confined to this component, where a correlation between two lines is otherwise undefined over unmatched features. The resulting neighbour lists inform substitution only; the score itself continues to withhold rather than impute (§3.3). The 2,000-feature cap applies only to RNA; the metabolomics and miRNA panels have fewer features. Variance filtering mitigates distance concentration on RNA, where dimensionality is highest.

The axes remain separate. Their top-25 neighbour lists overlap by only 7–15% and cover different line sets: of 1,673 lines with RNA, 726 have no other axis, so a merged score would reflect only RNA for those lines while losing track of that fact.

One safeguard is essential. Two nearly identical lines may be biologically related or cross-contaminated, meaning they are actually the same cells, and recommending these as alternatives

would falsely suggest independent replication. Candidates are checked against the Cellosaurus problematic-line register [7]; flagged lines are excluded or explicitly warned against.

The similarity metric was evaluated against an external criterion (§4.2.3). It is exposed both programmatically and through the interface, with the contamination check of the preceding paragraph applied to every returned candidate.

3.6 Agentic Interpretation and Provenance

The agent supplies interpretation and provenance after the ranking is established, without altering it. It is an interpretation layer positioned downstream of the deterministic pipeline, not an additional scoring model, and the architecture enforces the sequence: the statistical pipeline produces a validated output, and only then does a bounded agent process it.

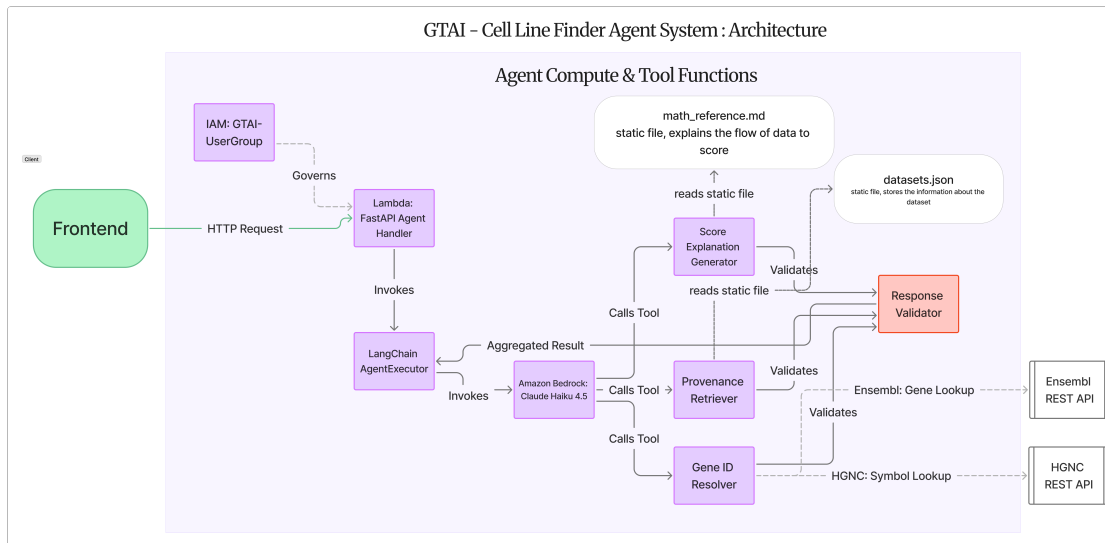


Figure 3.7: Agent architecture: client, compute and governance, orchestration, tools, and external data sources.

In the five-tier architecture of Figure 3.7, client requests reach an AWS Lambda function directly via a Lambda Function URL, with access restricted through an IAM action-allowlist policy. LangChain’s Agent Executor binds the available tools to the model and manages single-step tool selection. Amazon Bedrock hosts Claude Haiku 4.5, chosen for its speed, cost efficiency, and reliable tool routing, since routing here constitutes a classification task rather than a generative one. The model runs at temperature 0.0 with a maximum of 2048 output tokens, which minimises sampling variability in content restatement without guaranteeing bit-identical outputs from a hosted service. A single endpoint serves all three use cases, dispatching by query parameters. The model receives the query alongside a single system prompt shared across all use cases, then selects a tool and issues a structured function call:

- **Identifier resolution:** an alias or symbol is mapped to a canonical identifier via the Ensembl or HGNC REST API.
- **Score explanation:** the mathematical reference document is read and the relevant pipeline stage explained. No raw multi-omic values are supplied.
- **Provenance retrieval:** structured records describing the source datasets are returned.

Tools are stateless and without dependencies, each returning structured JSON for direct interface use. Outputs are validated to use only retrieved material, avoiding model-generated content, in line with the fourth design principle (§3.3): decomposed scores can be explained by retrieval, whereas generated explanations may diverge from the computation.

Limitations. Explanations depend on reference documents, so mismatches with the implementation may go unnoticed. The validator checks that retrieved material is used, not that the pipeline account is accurate. Identifier resolution depends on Ensembl and HGNC service state. Determinism holds for identical inputs, but outputs may vary as services or documents update. §4.3 evaluates robustness and deviations.

3.7 User Interface

The interface is organised around researcher actions rather than system components. The six requirements derive from the aims of §1.2, not from user research: each expresses a framework feature separate reporting of prioritisation and confidence (O3), three query modes (O4), and access to evidence and provenance (O1, O6). These are design choices rooted in objectives, not in observed user behaviour. No user study was performed; §4.4 checks only whether the interface displays validated output. The interface was designed around six user requirements:

Table 3.3: User requirements and the interface features answering them.

Requirement	Implementation
Identify suitable models quickly	Ranked table with normalised score bars and exact values
Query flexibly: target, selectivity or combination	Type-ahead query builder; mode inferred from input
Understand why a model scores as it does	Seven-step read-only scoring walkthrough
Assess confidence, not rank alone	Confidence tier, limiting gene, partial-evidence flags
Compare tissue contexts	Lineage grouping, recommendation and filter
Inspect molecular evidence	Per-line drill-down: rank, score, tier, alteration evidence, alternatives

A typical session moves from formulating a query to evaluating ranked candidates and reviewing a scoring walkthrough for a gene–cell-line pair. This read-only walkthrough links directly to the computational pipeline; each of its seven stages robust z -scoring, within-lineage RNA percentile, RNA-adjusted protein residual, cross-modality weights, combined core score, gene-role direction, and confidence level shows its formula and parameters. Only the explanatory prose is model-generated, ensuring alignment with the implemented algorithm. The application is a single-page React/TypeScript client. Queries route to FastAPI endpoints for the three query modes (§3.5), with responses passing through a typed adapter that separates UI and API. Appendix A.2.7 documents build tooling, styling, result visualisations, and request cancellation. Two design principles guide the interface: progressive disclosure shows the ranked table first, revealing visualisations, evidence, and methodology only on request to avoid information overload; *honest representation* normalises bar lengths, shows exact values, and reports confidence tiers, limiting genes, and partial-evidence flags to clarify ranking reliability. Whether these choices aid decision-making remains untested (§4.4).

Chapter 4

Critical Evaluation

Each of the four layers described in Chapter 3 is vulnerable to a different kind of failure, and each is therefore evaluated against its own question: whether harmonisation preserves information (§4.1); whether the statistical model computes accurately, remains stable, and produces recommendations consistent with external evidence (§4.2); whether the agentic layer explains without modifying (§4.3); and whether the interface faithfully presents validated output (§4.4).

4.1 Harmonisation Evaluation

Integration must preserve information. Chapter 2 identified five obstacles: entity identity, incompatible scales, structured missingness, version drift, and coverage asymmetry. Each was addressed during harmonisation or by a downstream component.

Entity identity was checked against independent references. Of 160,096 fusion partner symbols, 157,507 agreed and 2,589 (1.6%) disagreed; 420 of 12,558 protein map entries (3.3%) matched no gene in the roster, but all proteomics columns had mapping info. Most fusion disagreements were due to provisional versus approved names, indicating annotation drift, not joining error. Although 3.3% of protein entries are unmatched, all measured proteins have mapping info, so no measurements were dropped. Incompatible measurement scales were handled by standardising each layer by its properties rather than forcing all onto a common scale.

Structured missingness was not imputed. Missing measurements are represented through detection-regime allocation, withholding unavailable evidence rather than zero-filling. The composite-score fallback treats absent protein evidence as neutral, not low (§3.3.4), preserving the distinction between weak and unavailable measurements. Version drift was addressed by release pinning. Coverage asymmetry was recorded: coverage ranges from 91.9% for genomic signatures to 17.6% for proteomics. This imbalance bounds the score: a line without proteomic data cannot reach the extremes available to corroborated two-layer lines (§3.3.4). Coverage is reported alongside the confidence tier through partial-evidence flags and source and platform counts (Table 3.3, §A.2.6). Two one-to-many relationships—model to sequencing profile and cell line to GEO sample—were retained as set-valued identifiers, keeping table-to-roster joins one-to-one. Ambiguous cases were duplicated and flagged, causing at most 0.08% row inflation (Appendix A.1). Row counts before and after identifier attachment account for all differences, with no unexplained changes.

Information is preserved: identity mapping is consistent with a small, well-characterised residual; incompatible scales are handled by measurement properties; missingness is represented, not imputed; versioning is pinned; coverage asymmetry is shown through confidence; and joins neither duplicate nor lose rows. The resulting corpus provides a traceable basis for the evidence model and evaluation in §4.2.

4.2 Model Evaluation

4.2.1 Computational correctness

Before any behavioural claim, the implementation must be shown to compute the outputs specified in Chapter 3. Four test surfaces establish this: a diagnostic suite covering transcriptomic source agreement, protein platform tiering and composite-score distribution; a held-out ranking harness evaluating predictions against independent drug-sensitivity ground truth; a manual biological sanity panel; and co-selection join tests verifying that the joint score reduces to the minimum of its constituent per-gene scores, that missing values do not propagate into false results, and that the join admits no partial matches. The three executable surfaces pass; the biological sanity panel is a qualitative review and returns no pass/fail outcome (Appendix A.3.1).

The implementation therefore performs as designed. Computational correctness does not imply biological correctness, which this subsection does not address.

4.2.2 Stability of Recommendations

Two cell lines may score similarly yet differ in measurement stability, so the point score alone is insufficient. A Monte Carlo layer, applied at evaluation, propagates empirical disagreement from three RNA and two protein platforms through the scoring chain (§3.3.4), reporting a 95% uncertainty interval and two rank-stability measures: top- k membership frequency (global and within tissue) and win-rate. Win-rate measures sensitivity to input error, not frequentist coverage.

The implementation matched the unperturbed score within 10^{-8} and was stable across sample sizes, win-rate varying by no more than 0.9 percentage points between 10^3 and 10^4 draws (Appendix A.3.2).

Ranking was stable under the assumptions tested. Independent $\pm 50\%$ noise variation preserved win-rate ranking (mean Spearman 0.91); correlated error gave 0.93 at $\rho = 0.25$ and 0.88 at $\rho = 0.50$. Some genes were more sensitive (0.80, 0.66), so robustness is not uniform.

Point scores often saturate near 1.0, so the ranking distribution rather than the raw score distinguishes top candidates (Figure 4.1a). Global and within-tissue stability answer different questions: a line may lead its tissue while facing more global competition (Figure 4.1b), which reflects competition density rather than uncertainty.

In two-layer lines, protein is the dominant source of Monte-Carlo uncertainty: approximately 91% of simulated score variance is attributed to protein and 9% to RNA (Appendix A.3.2). Protein drives this uncertainty because it enters. RNA mainly affects between-line discrimination, while protein drives uncertainty because it enters as a volatile residual at the combination boundary, whereas RNA enters as a winsorised, standardised term (§3.3.4).

A second instability is structural: about two-thirds of two-layer lines switched combination variant at least once, toggling between the residualised and direct paths when the two evidence disagree (§3.3.4). These lines sit near a decision boundary where small measurement differences shift the rule. The interval does not separate structural instability from measurement noise; a per-line variant-switch rate would help but is not yet available.

The uncertainty model is limited: only observation-level RNA and protein uncertainty is propagated, while platform calibration, batch effects and scoring-constant uncertainty (β_g , ρ_g , dispersions, shrinkage) are held constant. Intervals may be too narrow where shared effects dominate. Stability here reflects insensitivity to measurement noise, not biological validity.

4.2.3 Biological and External Evaluation

Whether recommendations align with external evidence is the central question. No accepted ground truth exists for cell-line selection (§2.6), so this section uses proxies: large-scale perturbation and drug-sensitivity screens from other laboratories. These provide associative, not

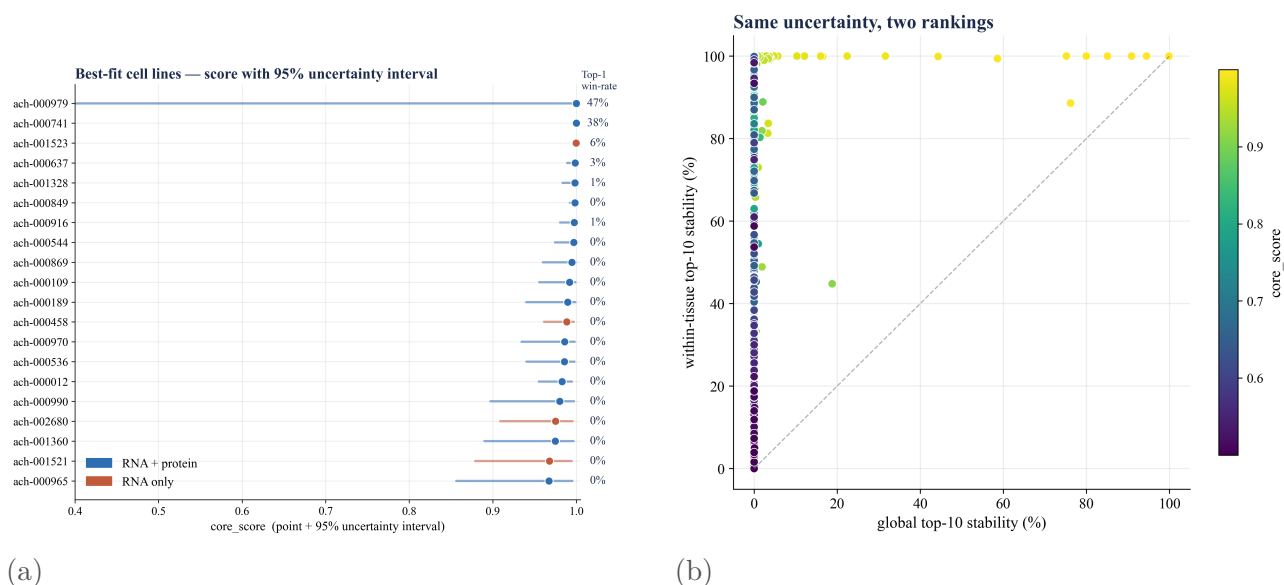


Figure 4.1: (a) 95% Monte-Carlo uncertainty intervals and top-1 win-rate for the twenty highest-scoring candidates on a single-target EGFR query, coloured by whether a line carries RNA and protein evidence or RNA alone. (b) Global versus within-tissue top-10 stability; mass above the diagonal marks lines stable within their tissue but contested globally.

validating, evidence. The composite score reflects relative target abundance and does not predict CRISPR dependency; where a line scores low for a gene that is also non-essential, the concordance suggests the score carries information beyond its design and supports its use as a prioritisation signal. It does not confirm any recommended cell line as biologically correct, prove causality, or show that abundance determines dependency. Each proxy has coverage limits and noise, addressed in the re-audit below.

Abundance–Dependency Relationship

The main audited claim is whether low transcript abundance predicts reduced CRISPR dependency [17, 68]. A ten-part re-audit was performed because a positive result can arise spuriously from the analysis pipeline.

The original broad claim was not supported. Nearly half of “not dependent” cell lines had not been screened by CRISPR and were default-labelled. Shuffling dependency labels reproduced 98% of the depletion effect, showing that label construction, not biology, drove the result. The original p -value ignored tissue clustering and was inflated by five orders of magnitude when clustering was considered [65].

A narrower effect remains after correction, concentrated among genes with less than 5% dependency prevalence, showing about 20–29% depletion and specific to CRISPR. An independent RNAi dataset [70] did not replicate this and showed the opposite trend. This CRISPR–RNAi divergence suggests a CRISPR-specific mechanism rather than abundance-dependency biology.

The leading candidate mechanism is cutting toxicity from guides targeting amplified regions, causing excess double-strand breaks independent of gene function [71]. Testing gave mixed results (Table 4.1): amplified lines showed more negative gene effect values, but no difference under a binary dependency threshold. Sparse, non-random copy-number annotation limits this test. A further limitation is that the shuffled placebo is itself significant in both comparisons, and in the opposite pattern to the observed results: the placebo permutes each gene’s amplification label independently across that gene’s own shared cell lines, preserving only its marginal amplified/non-amplified count and destroying both the true cross-gene chromosomal clustering of copy-number events and, since lineage is not represented in the permutation, any tissue-level

clustering; because it is therefore a looser null than the real data’s structure, these two p -values calibrate the test rather than support it. On this evidence the residual signal remains consistent with a copy-number artifact, but Table 4.1 should not be read as establishing the mechanism.

Table 4.1: Copy-number amplification and CRISPR gene effect.

Metric	Amp.	Non-amp.	Comp.
Median CRISPR gene effect	−0.085	−0.020	$\Delta = -0.045$, $p = 5.3 \times 10^{-12}$ (shuffled placebo $p < 0.001$)
Median dependency rate (threshold)	0.375	0.333	$p = 0.86$ (shuffled placebo $p = 0.010$)

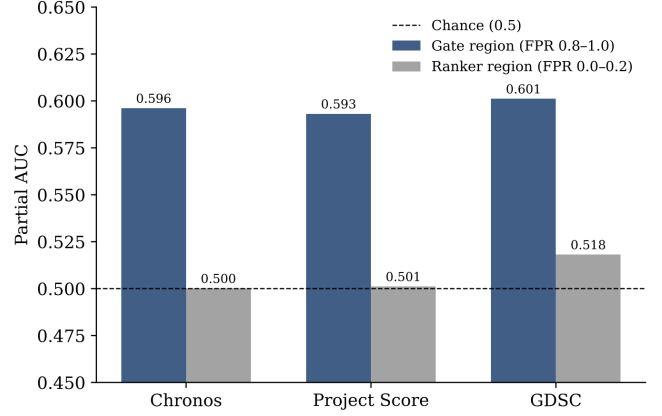


Figure 4.2: Partial AUC in the gate region and the ranker region across three label sources, two of them CRISPR screens [66].

Three label sources were used, two of them CRISPR screens that are not independent of one another: Chronos gene-effect estimates [68] (DepMap CRISPR), Sanger Project Score CRISPR screens [69], and GDSC drug-sensitivity labels [10]. The gate region (FPR 0.8–1.0) measures how well the score excludes candidates; the ranker region (FPR 0.0–0.2) measures how well it orders those admitted. The gate region performs above chance (0.5) for all sources; the ranker region is at chance for Chronos (0.500) and Project Score (0.501), and slightly above for GDSC (0.518), with no interval to judge significance (Figure 4.2). Thus, the layer acts as a gate, not a ranker: it identifies genes a line is unlikely to depend on, but does not order those it admits. The direction is consistent across all three label sources, but no interval is reported for any of the partial-AUC estimates, so the magnitude is not quantified; the cause of the residual signal is unclear.

Proteomic Evidence

An earlier assessment using global AUROC characterised the proteomic layer as a net negative. Re-examining the same comparison under partial AUC the portion of the metric reflecting performance at the top of a ranking reverses that conclusion (Table 4.2).

Table 4.2: Metric-dependent evaluation of the proteomic evidence.

Layer	GDSC pAUC	Chronos pAUC	Global AUROC
RNA alone	0.6415	0.6133	baseline
Production (RNA + protein residual)	0.6649	0.6304	−0.004 to −0.009
Difference	+0.023	+0.017	cost, not benefit

Partial AUC is reported for each label source; the Global AUROC column shows change relative to the RNA-only baseline, and the range −0.004 to −0.009 covers the same two label sources, not a parameter sweep. The metric was changed after the first assessment, so the

justification must stand on its own rather than on the outcome. The system is used as a top- N recommender, its output is consumed at the head of the ranking, and partial AUC measures performance in exactly that region, whereas global AUROC averages over regions no user inspects. On that argument partial AUC is appropriate here and global AUROC is not; the reversal of the earlier conclusion follows from the metric, and both are reported so the reader can weigh it. The protein evidence performs better under partial AUC, with only a minor drop in global AUROC. No interval is reported for either difference. It remains unverified whether high-tier recommendations are more often correct than moderate ones. The confidence tiers in §3.4 are not validated against recommendation accuracy, and no such claim is made.

Driver-Alteration Corroboration

Chapter 3 promotes lines carrying a corroborating driver alteration through the confidence tier rather than the score. Whether this improves practical usefulness was tested by measuring the rate at which known drug-sensitive lines [10] reach the top twenty [10] reach the top twenty (Table 4.3). Two estimands are reported: E1 weights every gene equally, and E2 weights genes by sensitivity.

Table 4.3: Driver-alteration corroboration: Hit@20 lift, with a sensitivity analysis excluding three genes flagged as fragile.

Estimand	Value	95% CI	p	n
E1, un-weighted mean	+0.0261	$[-0.003, +0.068]$	0.198	15
E2, sensitivity-weighted	+0.0132	$[-0.001, +0.030]$	0.093	15
E1, excl. 3 fragile genes	+0.0359	$[-0.000, +0.090]$	0.104	12
E2, excl. 3 fragile genes	+0.0227	$[+0.005, +0.042]$	0.057	12

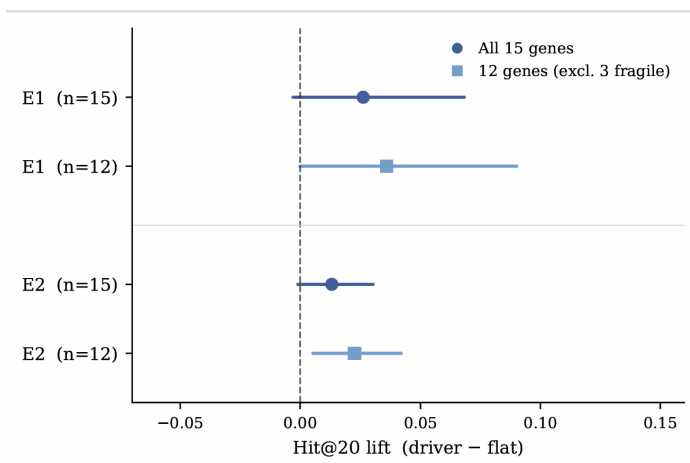


Figure 4.3: Hit@20 lift for both estimands was computed across fifteen role-annotated genes (E1 +0.0261, E2 +0.0132), and for twelve genes after excluding fragile ones (FGFR2, PIM1, MAP2K1: E1 +0.0359, E2 +0.0227). Error bars show 95% confidence intervals; only the twelve-gene E2 interval excludes zero.

All four estimates are positive, but none is statistically significant. In the full fifteen-gene sample, both intervals include zero ($p = 0.198$ for E1, $p = 0.093$ for E2). Excluding three genes flagged as fragile — FGFR2, PIM1, and MAP2K1, each of which hangs on a single-hit margin at rank 20 against a pool of well over 100 drug-sensitive lines increases both estimates, to +0.0359 for E1 ($p = 0.104$) and +0.0227 for E2. Only the latter’s interval excludes zero ($[+0.005, +0.042]$), while its permutation p-value of 0.057 does not; interval and p-value come from different procedures, and their disagreement at this sample size is itself a reason to treat the estimate as directional rather than confirmed. The exclusion was made after the full-sample result was seen, so the twelve-gene estimates are a sensitivity analysis, not the primary result.

The conclusion is weaker than intended. The evidence consistently suggests a positive association between genomic corroboration and prioritising drug-sensitive lines, even after excluding fragile genes, but no estimate is statistically supported. The confidence-tier promotion rule is not shown to improve usefulness, though it also does not appear to harm it in this small sample. Whether the association generalises, or any promoted line is a better model, remains untested.

Similarity Validation

The similarity graph was tested against a criterion independent of its own construction: whether lines it identifies as similar respond similarly to the same drugs [10]. Across 649 cell lines and 150 compounds it outperforms a same-tissue baseline by +0.186 [+0.149, +0.227], cluster-bootstrapped over cell lines. The advantage strengthens when the comparison is restricted to lines within the same lineage, which makes tissue of origin an unlikely sole explanation, though it does not exclude residual subtype structure shared by lineage and drug response. The metric is validated against this external criterion and is reachable both programmatically and through the interface (§3.5.1).

Evidence-Ranking Query Layer

Multi-gene queries are resolved using a three-valued logic that returns yes, no, or explicit abstention. Against a manually curated set of gene-line pairs, the layer aligns with the curated truth in just over 80% of cases where it gives a definitive answer. Where it does not commit, the system abstains for one of four specified reasons (Appendix A.3.3), operationalising the principle that absence of evidence does not constitute evidence of absence.

4.3 Agentic AI Evaluation

The agent must retrieve, interpret and explain model results without modifying or fabricating them. The architectural boundary is fixed: the statistical model generates the ranking upstream, and the agent does not recalculate, alter or override it. The agent is therefore evaluated only on retrieval, interpretation and explanation (Table 4.4; per-module detail in Appendix A.3.4). Tier 1 shows that the tools do not return fabricated numeric values, so a fabricated *score* cannot originate at the tool layer; this constrains where hallucination can enter but does not prevent unsupported statements in the surrounding prose. Tier 3, run against the deployed system, verifies correct tool selection across phrasings and confirms that fabricated gene-cell-line pairs prompt a methodology explanation rather than a fabricated score. Two kinds of failure were identified. First, a formal test failure: when both nomenclature services were unreachable, the local fallback failed on an empty index. This was the only failed assertion among 61 tests, showing that graceful degradation is not guaranteed when external dependencies fail. Second, an explanation-fidelity failure: in a DepMap provenance query the model asserted "three RNA expression sources". The statement is true of the corpus (§3.2) but was not supported by the tool output it purported to describe, so the failure is one of grounding rather than of fact the harder case to detect (Appendix A.3.4). The suite does not currently assess explanation fidelity. Elsewhere the model correctly expanded LFS1 to Li-Fraumeni Syndrome 1, showing that similar elaboration can be accurate, which makes unsupported explanations difficult to suppress without also removing useful ones. Neither failure breaches the architectural boundary: the agent never altered or overrode a ranking in any test, and the architecture gives it no path by which to do so. Unsupported explanations reduce explanation quality without affecting recommendations. The agent is nonetheless not immune to hallucination, so provenance statements should not be accepted at face value. No adversarial testing of the boundary was performed, so the separation rests on the architecture rather than on evidence that it resists attempts to circumvent it.

Table 4.4: Agent test suite by tier.

Module	Tier	Tests	Result
test_dataset_info	1	11	11 passed
test_score_explainer	1	10	10 passed
test_gene_alias	1 / 3	10	9 passed, 1 failed
test_api	2	15	15 passed
test_latency	2	8	8 passed
test_tool	3	1	1 passed
test_agent_lambda	3	6	6 passed

4.4 User Interface Evaluation

The final layer must present validated data accurately. No formal usability evaluation was performed, so there are no data or claims about usability or user satisfaction. Evaluation is limited to whether the interface accurately represents the pipeline’s output, an intrinsic software property. Three interface properties were verified. The scoring walkthrough shows formulae and parameters directly from the pipeline, maintaining quantitative consistency (§4.3). Tier 2 tests confirmed structured output is parsed correctly, reverting to prose when needed. Uncertainty and abstention are shown explicitly: confidence tier, limiting gene, and partial-evidence indicators appear with each result; abstentions are displayed; and score bars are normalised to avoid overstating differences, which also prevents the clustering of scores near the upper bound (§3.3.4) from being read as close agreement between candidates. The alternatives view exposes the similarity component of §3.5.1, with the contamination check applied to every candidate it returns. The interface displays model outputs, including uncertainty and refusals, without modification. Whether this aids researcher decisions is untested and remains a limitation.

4.5 Summary

Evidence strength differs by layer. Harmonisation preserves information: identity mapping is consistent, missingness is represented, and joins are lossless, though coverage is uneven. The model works as designed: scores behave as intended, and rankings are stable under uncertainty. However, protein dominates uncertainty in two-layer lines; two-thirds are near an unstable boundary and calibration, batch effects, and scoring constants are not propagated.

Biological evidence is the weakest layer because every result relies on external labels from other laboratories, none of which is treated as ground truth. Abundance acts as a gate, not a dependency ranker. Proteomic evidence performs better under partial AUC, the metric aligned with top- N recommendation. The evaluation removed one claim and narrowed a second. The abundance–dependency finding was 98% reproducible under shuffled labels, with significance overstated by five orders of magnitude. Driver-alteration lift is positive across all four estimands but not statistically supported: E1 +0.0261 ($p = 0.198$) and E2 +0.0132 ($p = 0.093$) across fifteen genes, rising to E1 +0.0359 ($p = 0.104$) and E2 +0.0227 ($p = 0.057$) in a post-hoc twelve-gene sensitivity analysis. Only the last interval excludes zero, and its permutation p -value does not meet conventional significance. The copy-number mechanism, the promotion rule and recommendation correctness remain open questions. Chapter 5 considers what follows.

The agentic layer never altered rankings in any test; one formal test failed, and one unsupported explanatory claim was observed, showing the architecture constrains error consequences without eliminating error.

Chapter 5

Conclusions

5.1 Contributions

Chapter 2 identified the gap: public cell-line molecular data exist, but converting it into auditable ranked recommendations with confidence, explanation, and alternatives remains manual and undocumented. The system here addresses that gap; methods and evidence are in Chapters 3 and 4. Key contributions: C1: Auditable multi-omic corpus: Nineteen public tables from seven programmes were unified into 2,127 cell lines and 20,163 protein-coding genes over nine measurement types, with only 1.6% fusion-symbol disagreement, no unmapped protein columns, and at most 0.08% row inflation (§4.1). Identity was resolved through stable accessions, critical given 580 names shared between human and non-human entries. C2: Layer-specific evidence modelling: Each layer is standardised by its measurement properties, not forced onto a common scale, and each layer’s standardisation is defined by what its measurements can support (§3.3). The genomic evidence is treated as a bounded ordering, not a calibrated probability. C3: Separable score and confidence: Score and confidence tier are computed and reported independently; candidates with alterations contradicting abundance are flagged, not excluded. This separation is functionally correct (§4.2.1) but not yet validated against recommendation accuracy (§4.2.3, §5.4). C4: Matched query semantics: Three query modes use distinct rules, with a structural guard preventing Y-linked genes from being rewarded as evidence of absence, and abstention for four recorded reasons where evidence is insufficient. Where the query layer commits to an answer, it agrees with curated labels in just over 80% of cases (§4.2.3). C5: Interpretation layer cannot alter results: The agentic layer resolves identifiers, retrieves provenance, and explains scores without changing rankings. One unsupported provenance claim was observed in a passing test (§4.3); thus, the true contribution is enforcing the boundary between interpretation and decision, not assuming error-free explanation.

5.2 Status Against Aims and Objectives

The following points summarise each objective from §1.2 and its supporting evidence. Six objectives were met:

1. Integrated heterogeneous public multi-omic sources into a queryable corpus with provenance, achieved via the harmonisation pipeline (1.6% fusion-symbol disagreement, no unmapped protein columns, at most 0.08% row inflation; §3.2, §4.1).
2. Modelled each evidence layer by its measurement properties, standardising each layer on its own terms rather than forcing a common scale (§3.3).
3. Produced ranked recommendations with confidence separated from score; this division is functionally correct (§3.3.4–§3.4, §4.2.1), though outcome validation is pending (tier–outcome correspondence is the next test; §4.2.3).

4. Supported target, co-selection, and exclusion queries through a query and ranking layer: conjunctive logic was unit-tested and agreed with curated truth in over 80% of cases (§3.5, §4.2.1, §4.2.3).

5. Quantified uncertainty in recommendations within a defined scope: stability analysis gave a Spearman correlation of 0.91 under $\pm 50\%$ parameter changes, with uncertainty propagated at the observation level and platform calibration, batch effects, and scoring constants fixed (§4.2.2).

6. Provided interpretation and provenance without altering the ranking: no boundary violation in any test, with 60 of 61 tests passing and one formal failure with a characterised fidelity observation (§4.3).

7. The similarity component also met its objective and is externally validated: the metric outperformed a same-tissue baseline across 649 cell lines and 150 compounds (§4.2.3). The component is reachable both programmatically and through the interface, with the cross-contamination check applied to returned candidates.

8. Biological usefulness is established as far as the evidence allows, a limit of the problem, not the work. No accepted ground truth exists for cell-line selection (Chapter 2); evaluations use proxies—CRISPR essentiality and drug-sensitivity data from other labs, each with its own noise and coverage biases (§4.2.3). Within these limits, the evaluation: (1) supports the claim that abundance evidence functions as a gate; (2) finds driver corroboration positive but not significant in a fifteen-gene sample; and (3) retracts a claim that did not survive audit. The framework distinguishes supported, suggestive, and unsupported states, making the supported claim more reliable.

5.3 Limitations

Limitations are summarised here for reference. The main limitation is the absence of biological ground truth: all external evaluations use proxies, and no recommendation is biologically confirmed (§4.2.3, §5.2). Confidence is not outcome-validated: tiers separate score from evidence by design, but have not been tested for correctness (§4.2.3). Uncertainty has limited scope: only observation-level error is propagated; calibration, batch effects, and scoring constants are fixed (§4.2.2). Margin is not reported as a confidence component, though identified as necessary (§2.6). Version drift is addressed by release pinning, not independently tested (§4.1). Documentation and implementation are not automatically reconciled: explanations are based on reference documents, so discrepancies may go undetected (§3.6). Agent explanation fidelity is imperfect: the agent cannot alter rankings, but unsupported explanatory claims are still possible (§4.3). No external baseline: the composite was compared against its own RNA-only arm, not against ranking by a single source in native units, so the benefit of harmonisation is not quantified (§4.5). Coverage bounds the attainable score: because absent proteomic evidence enters as a neutral value under an unchanged denominator, lines without proteomic measurement cannot reach the extremes available to corroborated two-layer lines (§3.3.4). No multiplicity control: per-gene concordance tests, correlations and lift estimates are computed across many genes without correction. Source count is not normalised: lines measured by more transcriptomic sources have more dispersed combined scores and are not directly comparable on the raw combined scale (Appendix A.2). Finally, there is no usability evidence: the interface was evaluated only for faithful exposure, with no user-centred evaluation or usability claims (§4.4).

5.4 Further Work

Several directions follow from unexpected findings:

CRISPR–RNA divergence: The abundance dependency effect appears in CRISPR but not RNAi data, and sometimes in the opposite direction (§4.2.3). If abundance predicted dependency biologically, both methods would show it. This suggests a CRISPR-specific mechanism, possibly cutting toxicity in amplified regions. Copy-number tests are suggestive, but sparse, non-random annotation limits resolution. Focusing on lines with dense copy-number data and guide-level effect estimates could resolve this and inform broader CRISPR screen interpretation.

Protein-term trade-off: In two-layer lines, protein accounts for about 91% of score variance under the Monte-Carlo model (§4.2.2), reflecting sensitivity to measurement noise rather than biology. Adding the RNA-residualised protein term raised partial AUC (Table 4.2; no interval reported) but introduced this volatility a trade-off not anticipated in Chapter 3. No residualised-versus-unresidualised comparison was run, so the trade-off is attributable to the protein term rather than to residualisation itself. Whether a shrunk residual or volatility-adjusted protein influence can retain the partial-AUC gain with lower variance remains to be tested.

Combination-variant instability: About two-thirds of two-layer lines changed combination variant in at least one Monte-Carlo draw, indicating proximity to a decision boundary rather than just noisy scores. Reporting a per-line regime-switch rate with the uncertainty interval would separate these; this was identified but not implemented. The same applies to *margin* (§2.6), which is computable but not reported.

Validating confidence tiers: The rule placing candidates with a directionally consistent driver alteration in the **HIGH** tier is untested against outcome: it is unknown whether **HIGH**-tier recommendations are more often correct than **MEDIUM**-tier ones (§4.2.3).

Uncertainty beyond observation level: The model holds scoring constants fixed. Propagating uncertainty in β_g , ρ_g , and shrinkage parameters, and modelling platform-level calibration, would widen intervals where shared effects dominate and test stability persistence.

Constraining agent provenance claims: The agent’s main failure was a provenance statement unsupported by tool output (§4.3). Since the same elaboration produced correct expansions elsewhere, suppressing it would remove useful explanation. Restricting factual claims to a closed vocabulary from the record, while leaving prose free, is preferable. The test suite should also check explanation fidelity, which it does not currently do.

User-centred evaluation: The interface was evaluated for output accuracy, not for decision support (§4.4). Whether researchers interpret the **CONTEXT** tier or handle abstentions properly remains an open empirical question for confidence machinery effectiveness.

5.5 Closing Remarks

This dissertation presents an auditable framework for target-specific cell-line prioritisation, in which every ranking position is traceable to the evidence that produced it. When evidence is missing, the score receives no evidence from it rather than an imputed value; when two sources disagree, it lowers confidence rather than selecting between them; and when a gene is structurally absent, it declines to score rather than assigning a score of zero.

The framework does not establish universal biological correctness and makes no claim regarding usability. An initially attractive headline result did not withstand audit, and the remaining claim is narrower and more useful: abundance evidence excludes candidates more consistently than it ranks those it admits. The framework’s ability to detect and constrain its own overclaims is not a weakness; it underpins the trustworthiness of the remaining findings.

Appendix A

Technical Supplement

This appendix records parameter values, resolution rules and diagnostics supporting Chapters 3 and 4. Conceptual explanations are not repeated here.

A.1 Harmonisation and Validation Details

Identifier resolution. Where a source row carries an RRID, the Cellosaurus accession is extracted from it to fill any gap; an existing accession is never overwritten. Cellosaurus synonyms are merged through the accession bridge, widening the name set available to the final matching route. Gene aliases are retained rather than collapsed, since one Ensembl gene legitimately carries several names through annotation drift. UniProt and COSMIC identifiers are linked through gene symbols, each reverse mapping generated by inverting its forward mapping rather than by matching a second time an independent reverse match could produce a mapping inconsistent with its own inverse.

Cardinality. Measured in both directions before aggregation; Table A.1 reports the four principal relationships.

Table A.1: Cardinality of the principal identifier relationships.

Relationship	Forward	Reverse	1:1
Profile → model	0 / 3,830	1,421 / 1,822	No
GEO sample → CVCL	0 / 3,267	419 / 797	No
CVCL → HPA line	0 / 1,198	0 / 1,206	Yes
CCLE name → model	0 / 928	0 / 927	Yes

The forward and reverse columns report the number of identifiers mapping to more than one counterpart, out of the total examined in that direction; **max** is the largest number of counterparts observed for a single identifier. The GEO relationship is the most skewed: a single cell line accounts for up to 478 samples, so replicate depth in that source is concentrated on a small number of heavily studied lines, and the $\sqrt{n}$ within-source weighting of §3.3.1 gives those lines a more precise combined score than lines with a single sample.

Scope filtering. The 580 human/non-human name collisions comprise 30 primary-to-primary (including **ca**, **ham-1**, **me1**, **mk2**), 220 human-synonym to non-human-primary, 134 the reverse, and 196 synonym-to-synonym. Mixed-origin models are retained if any species field indicates human origin; blank species fields are excluded and counted separately; an absent species column causes every row to be dropped with a warning rather than passing through unfiltered.

The protein-coding whitelist was inferred as the union of the filtered mutation table (18,103 genes, undercounting because it lists only genes both coding and mutated somewhere) and `hpa_rna` (20,162 genes, independent of mutation status). Their union of 20,163 approaches the $\sim 19,500$ protein-coding genes estimated for the human genome, which is the principal evidence the inference is sound. Applying it retained 19,897 of 53,961 DepMap expression columns and 16,063 of 19,914 GEO columns; fusion events were reduced from 184,237 to 80,048 across 1,698 cell lines, giving the 160,096 partner symbols audited in §4.1.

Ambiguity volumes. Duplication affected `cellosaurus` (4 rows), `hpa_desc` (1), `geo_expr` (2) and `hpa_rna` (20,162 rows from 40,324 ambiguous rows, an inflation of 0.08%).

Validation outcomes. Fusion partner symbols were checked against reference names for the same gene, with the reference set *excluding* fusions—the roster’s names are built partly from fusion data, so comparing against them would be circular. Of 160,096 symbols, 157,507 agreed and 2,589 (1.6%) did not, overwhelmingly clone-based provisional names against approved symbols: annotation-vintage drift rather than incorrect joins. On the protein side, 420 of 12,558 map entries (3.3%) have a symbol matching no gene in the roster, but *zero* proteomics columns are absent from the map the second figure being the important one, since it establishes that every measured protein has mapping information. Cross-source gene-name checks flag only genes where two sources share *no* name, since a gene mapping to several names is normal.

A.2 Statistical Model Parameters

A.2.1 Method resolution rules

Processing family is recovered from the free-text `!Sample_data_processing` field of each SOFT record. Rules are applied in the order shown and the first match wins; order is significant for gcRMA before RMA (substring) and for MAS5, whose alternatives capture environment or operation names used in place of the algorithm. Unmatched samples take family `unknown`; samples with no record take method `UNKNOWN|unknown`. Both are retained and flagged. Table A.2 lists the rules in order. Five of the ten families are observed in the data used here; the remainder are retained so that the rule set is complete against the platforms GEO exposes, and unobserved families cost nothing at match time.

GEO coverage audit: all 3,267 samples in `geo_expr` carry a metadata record; of these, 81 could not be resolved to a named processing family and take the residual `unknown` family. The seventh method, `UNKNOWN|unknown`, is reserved for samples with no record at all and is unpopulated in this compendium, so the six rows of Table A.3 account for every sample.

A.2.2 Transcriptomic estimator

RNA is standardised within each (gene, lineage) stratum, as set out in §3.3.1. The stratum index L in Equation (A.1) denotes lineage; n_L is the number of cell lines in that lineage carrying a finite value for the gene; and the shrinkage pulls the dispersion estimate of a small lineage toward the gene-global estimate $\hat{\sigma}_g$, so that a lineage with too few lines to support its own scale borrows one rather than producing a defined but unstable estimate:

$$w_L = \frac{n_L}{n_L + n_0}, \quad \tilde{\sigma}_{L,g} = \max\left(\sqrt{w_L \hat{\sigma}_{L,g}^2 + (1 - w_L) \hat{\sigma}_g^2}, 0.1 \hat{\sigma}_g, \varepsilon\right). \quad (\text{A.1})$$

The lower bounds in Equation (A.1) guarantee a positive divisor where a lineage’s values tie at the detection floor and its median absolute deviation is zero. Scale provenance is recorded with every score as lineage, shrunk or fallback_global, so any borrowed scale is identifiable in a query.

Table A.2: Ordered lexical rules. The scale column is a prior only; the empirical detector decides.

#	Family	Pattern	Prior
1	gcrma	\bgc [\s \-]?rma\b	log
2	rma	\brma \b	log
		robust\s +multi[\s \-]?array	
3	mas5	\bmas \s *5 \bgcos \b	linear
		microarray\s +suite\s *5	
		trimmed\s +mean.*target	
		global\s +scaling	
4	plier	\bplier \b	log
5	dchip	\bdchip \b	linear
		model[\s \-]based\s +expression	
6	mas4	\bmas \s *4	linear
		average\s +difference	
7	vsn	\bvsn \b	log
		variance\s +stabilis?z?ing	
8	quantile	quantile\s +normal	unknown
9	lowess	\blowess \b \bloess \b	log
10	rnaseq	\brsem \b \btpm \b	linear
		\bfpk \b \brpk \b	
		\bdeseq \b	

Table A.3: Per-method scales, floors, and sample counts from `geo_methods.csv`.

Platform	Method	Scale	n_{GSM}
GPL570	rma	log	1523
GPL570	gcrma	log	951
GPL570	mas5	linear	471
GPL14877	gcrma	log	178
GPL570	unknown	unknown	81
GPL570	plier	log	63

The dispersion of the combined score also varies systematically with the number of contributing sources measured as 0.88, 1.54 and 1.86 for one, two and three sources. This is the expected consequence of combining positively correlated sources: the inverse-normal combination below assumes independent inputs, so a score formed from three correlated transcriptomic sources ($\bar{\rho} = 0.548$, §A.2.5) is over-dispersed relative to one formed from a single source, and a line measured by all three is not directly comparable on the raw combined scale to a line measured by one. The source count k_{src} is recorded with every score (§A.2.6).

Regime selection uses the detection fraction $d_{L,g}$, the proportion of finite values at or above the floor: $d \geq q_{\text{upper}}$ takes the robust z ; $q_{\text{lower}} \leq d < q_{\text{upper}}$ takes $z_{g,c} = \Phi^{-1}((R_{g,c} - 0.5)/n_{\text{fin}})$; below q_{lower} , and where no finite value exists, no score is emitted. Combination uses $Z = \sum_i w_i z_i / \sqrt{\sum_i w_i^2}$, with $\sqrt{n}$ weights within source and equal weights across sources. Because the regime is chosen per gene and lineage, the two estimators are calibrated within a gene rather than across genes.

A.2.3 Calibration

Exhaustive grid search subject to $q_{\text{lower}} < q_{\text{upper}}$; since every candidate lower value is below every candidate upper value the constraint removes nothing and the grid comprises $4 \times 5 \times 6 \times 4 = 480$ settings. Calibration uses a seeded sample of 1,200 genes, scoring a setting only where at least 200 comparable observations fall in the evaluation band $0.01 < d < 0.90$. The held-out reference is cached under fixed parameters $q_{\text{floor}} = 0.10$, $n_0 = 25$, $q_{\text{upper}} = 0.50$, $q_{\text{lower}} = 0.10$, so the target cannot move with the candidate being scored. Selection applies the one-standard-error rule, taking the most parsimonious setting within one SE (Table A.4)—highest q_{upper} , then q_{lower} , then n_0 , each direction favouring weaker distributional assumptions.

The final row indicates identifiability: a small spread would mean the data do not determine n_0 , bounding the weight the selected value should carry. The maximum concordance attained is small in absolute terms ($\rho^* = 0.0578$) and ten of the 480 settings fall within one standard error of it, so the criterion separates settings weakly. The one-standard-error rule therefore

Table A.4: Calibration grid and outcome.

Parameter	Candidates	Selected
q_{floor}	0.02, 0.05, 0.10, 0.20	0.2
n_0	5, 10, 25, 50, 100	100
q_{upper}	0.30, 0.40, 0.50, 0.60, 0.70, 0.80	0.60
q_{lower}	0.02, 0.05, 0.10, 0.20	0.2
Maximum concordance ρ^* / SE		0.0578 / 0.0007
Settings within one SE		10
Spread of best ρ across n_0		0.0002

selects on parsimony rather than on a sharp optimum, and the chosen values should be read as a defensible default rather than a tuned optimum. The concordance criterion is internal to the transcriptomic layer and is unrelated to the external labels used in §4.2.3, so tuning here does not contaminate the biological evaluation.

A.2.4 Genomic channel parameters

Table A.5 gives the six channels and their Hill parameters.

Table A.5: Genomic evidence channels and Hill parameters.

Layer	Channel	Input	p_0, k
Mutation	Consequence	VEP impact rank [53]	2.5, 2.0
Mutation	Pathogenicity	AlphaMissense [54] / CADD [55]	0.5, 2.0
Mutation	Burden	Variant count, censored at five	3.0, 1.5
Fusion	Caller confidence	Ordinal low/medium/high	2.0, 1.5
Fusion	Expression support	Best FFPM [31]	0.5, 2.0
Fusion	Recurrence	Samples observed	2.0, 2.0

The in-frame increment is $b = 0.15$, applied additively after combination [58]. Copy-number flags are alteration = (amp $\wedge$ $r_g \in \{\text{oncogene, both}\}) \vee (\text{del} \wedge r_g \in \{\text{tsg, both}\})$, with all remaining called events as context; the two are mutually exclusive and jointly exhaustive. Categorical COSMIC calls are used rather than thresholded dosage, since COSMIC applies a higher amplification threshold to genome-duplicated samples, defining duplication as average ploidy above 2.7 [59]. Multiple observations per gene and cell line are reduced by the maximum; copy-number flags by disjunction.

A.2.5 Composite score parameters

Per-gene RNA–protein correlation is shrunk toward a panel-wide prior by a fixed-weight linear rule of the James–Stein type [64], and the slope formed from it:

$$\rho_g = \lambda \rho_0 + (1 - \lambda) \hat{\rho}_g, \quad \lambda = \frac{K}{n_g + K}; \quad \beta_g = \rho_g \cdot \frac{\tilde{\sigma}_g^{\text{prot}}}{\tilde{\sigma}_g^{\text{RNA}}}, \quad (\text{A.2})$$

with $\rho_0 = 0.353$, $K = 30$ and $N_{\text{min}} = 30$ shared cell lines required before ρ_g is estimated. The shrinkage constant K is fixed by convention rather than estimated from data, so λ is a pre-set weight rather than a data-derived empirical-Bayes strength. Using ρ_g directly as a slope would over-correct by the dispersion ratio (1.525 for RNA against 0.951 for protein).

Layer weights derive from the Kish effective sample size, $n_{\text{eff}} = N/(1 + (N - 1)\bar{\rho})$: three transcriptomic sources ($\bar{\rho} = 0.548$) give 1.431, two proteomic platforms ($\bar{\rho} = 0.373$) give 1.45; each layer is weighted by $\sqrt{n_{\text{eff}}}$. The weights therefore express how much independent evidence each layer contributes, not how precisely it is measured; the consequences for uncertainty are

reported in §4.2.2. Where both layers are confidently signalled ($|z| > 1$) but point in opposite directions, the composite is formed from the standardised layers directly without residualisation, with the variant recorded in an audit column.

Because the standardised layers entering Equation (3.5) are not exactly unit-variance, the mapped scores concentrate near the upper bound for strongly supported candidates; this is the saturation reported in §4.2.2.

Tier assignment uses $\tau = 0.80$; directional consistency is $\pi_{g,c} \leq 1 - \tau$ for loss-of-function alterations and $\pi_{g,c} \geq \tau$ otherwise, with dual-role genes treated as gain-of-function. A candidate is *leading* where $\pi_{g,c} \geq \tau$. The four tiers form a complete partition, verified at write time by assertions on their defining predicates.

A.2.6 Output schema

`transcriptomics_z` is written gene-major so the storage layer can skip row groups when filtering on gene; at 22,045,461 rows it is held in DuckDB. Columns: `ensg`, `model_id`, `lineage`, `z_lineage` (primary), `z_global` (secondary), `k_src`, `k_methods`, `det_frac`, `status`, `scale_source`, and per-source scores `z_hpa_rna`, `z_depmap`, `z_geo`. Corpus totals: $\approx 20,000$ genes and $\approx 2,000$ cell lines.

`protein_z` carries the adjusted score, the pre-attenuation score, κ_g , the platform count, and the isoform count, with per-gene isoform decisions in `protein_isoform_qc`. The platform count and pre-attenuation score together distinguish a gene attenuated to zero by cross-platform disagreement from one with no proteomic measurement at all, a distinction the composite itself does not carry (§3.3.2).

A.2.7 Interface implementation

The single-page React and TypeScript client is built with Vite and styled with Tailwind CSS. It comprises a query builder with type-ahead gene search and a results-limit control; a ranked results view with a switcher between a ranked table showing global and within-lineage scores, an evidence grid displaying cell lines against their multi-omics evidence layers for each gene, and a lineage breakdown; a per-line cell-line profile; and a scoring-methodology panel. The similarity component of §3.5.1 is surfaced through the alternatives view. A typed adapter layer isolates the application from the API, with queries routed by target count and the presence of an exclusion to the four ranking endpoints of the FastAPI service, (`/prod/gene`, `/prod/genes`, `/prod/exclude/many`, and `/prod/genes/exclude/many`), corresponding to the single-gene, joint multi-gene, selectivity, and joint-selectivity modes described in §3.5. Supporting endpoints provide per-line detail, lineage-grouped variants, and the gene and cell-line reference data used for autocomplete and lookup. Requests are handled asynchronously, with superseded requests cancelled and responses parsed into strongly typed domain objects.

A.3 Evaluation Supplement

Supporting detail for Chapter 4. Results are interpreted there; this section records test coverage and verification outputs.

A.3.1 Functional test coverage

Four test surfaces support the functional claims of §4.2.1; three are executable and return pass/fail outcomes, the fourth is a qualitative review.

- **Diagnostic suite.** Source agreement in the transcriptomic layer, protein platform tiering, and the shape of the composite score distribution, alongside a held-out evaluation summary reporting the sign of the driver-alteration lift across the three gene-role classes under the

current pipeline state. The summary records direction only; the estimates and their intervals are given in §4.2.3, where no estimate reaches significance.

- **Held-out ranking harness.** Scores predictions against independent drug-sensitivity labels; provides the substrate for the gate audit (§4.2.3) and the alteration result (§4.2.3).
- **Co-selection join tests.** Executed unit suite confirming that the joint score reduces to the minimum of constituent per-gene scores, that missing values never silently propagate into a false result, and that the join is strictly conjunctive rather than admitting partial matches: a candidate lacking a score for any requested gene is not returned.
- **Manual biological sanity panel.** Qualitative review of named genes and cell lines against known tissue-restricted expression patterns. This surface returns no pass/fail outcome and is reported as a review rather than a test.

A.3.2 Uncertainty layer verification

Table A.6 records the seven verification and robustness checks.

Table A.6: Verification and robustness checks for the Monte-Carlo uncertainty layer, run against the pipeline over a thirty-gene sample.

Check	What it tests	Result
Centre-draw identity	Un-perturbed recompute equals the stored core score	$\max \Delta \approx 1.8 \times 10^{-8}$ (EGFR, TP53, MYC)
Order-consistency	Disjoint-interval pairs keep their order across draws	100% of 570 pairs order-stable at $\geq 95\%$
Self-consistency	Draw median versus stored centre, by score bin	mean $ \text{median} - \text{centre} = 0.006$
Convergence	Estimates stable as N grows	win-rate change ≤ 0.9 pp, $N = 10^3 \rightarrow 10^4$
Knob robustness	Win-rate ranking under $\pm 50\%$ parameter changes	mean Spearman 0.91
Variance attribution	RNA versus protein share of score variance	two-layer lines: protein $\approx 91\%$, RNA $\approx 9\%$
Independence sensitivity	Win-rate ranking under correlated error	mean Spearman 0.93 ($\rho = 0.25$), 0.88 ($\rho = 0.50$)

Self-consistency detail. The difference between draw median and stored centre was near zero at mid-range scores and small and directional near the bounds—positive near 0, negative near 1—which is the expected signature of the compression of Φ near its limits.

Variance attribution method. Each two-layer line’s score variance was decomposed by re-running the simulation three times: perturbing RNA alone, protein alone, and both. Writing V_R , V_P and V_{both} for the resulting variances, the first-order contributions are V_R and V_P and the interaction is $V_{\text{both}} - V_R - V_P$. This is a variance-based sensitivity decomposition in the sense of Sobol’ [67], not a full Sobol’ index computation, which would require a specific independent-input estimator; it is therefore reported as an approximate first-order attribution. Normalising first-order terms to sum to 100%, protein accounted for approximately 91% and RNA approximately 9% across two-layer lines (protein 80–100% across genes), with the interaction term of order 3% of the total. The decomposition is defined only where both layers are present, so it does not extend to the single-layer lines that make up most of the corpus; those lines have no protein term to perturb and consequently narrower intervals (§4.2.2).

Independence sensitivity detail. Perturbations were re-drawn with positive error correlation across a twenty-gene sample. Largest single-line win-rate shifts averaged 1.4 and 2.7 percentage points at $\rho = 0.25$ and $\rho = 0.50$ respectively. The effect scaled smoothly with assumed correlation and was small for most genes, though a minority were more sensitive: minimum Spearman 0.80 and 0.66 at the two correlation levels. This varies observation-level error draws only; correlation induced through the estimated constants, which are held fixed, is outside scope.

A.3.3 Query-layer abstention conditions

The three-valued evidence-ranking layer of §4.2.3 returns an explicit abstention rather than a guess under four named conditions:

- the cell line is entirely absent from the relevant expression dataset;
- the measured value sits within the uncertainty band around the detection threshold, too close to call;
- the line was never sequenced for the assay in question; or
- only variants of ambiguous functional impact are available, for which the underlying variant-effect predictor cannot supply a confident severity ranking.

Each abstention is recorded with its reason, so a downstream consumer can distinguish an unmeasured pair from a measured but borderline one.

A.3.4 Agent test-case detail

Per-module findings supporting the tier summary of §4.3.

Configuration. The scoring-explanation agent uses Claude Haiku 4.5 via AWS Bedrock (profile `eu.anthropic.claude-haiku-4-5-20251001-v1:0`, region `eu-west-2`). The model runs with temperature 0.0 and `max_tokens` 2048, which minimises sampling variability in content restatement. Orchestration uses LangChain (`langchain==1.3.14`, `langchain-core==1.5.3`, `langchain-aws==1.7.2`); `langchain-aws` integrates LangChain’s tool-calling with Bedrock’s Converse API for structured tool use. A single shared system prompt is assembled at startup and routed conditionally by use case, so the three use cases of §3.6 differ in the routing branch taken rather than in the prompt loaded. The prompt has three components: (i) role and routing logic, (ii) embedded mathematical reference, (iii) structured-output contract. The mathematical reference is version-controlled; results here use commit `db5255b`, and reproducibility depends on pinning this value. The agent is containerised with Docker and deployed to AWS Lambda. As noted in §3.6, outputs may drift as models or references update; the configuration pinned here matches the reported results, and a re-run on an updated model or prompt should reproduce equivalent, though not necessarily identical, output.

test_dataset_info.py (Tier 1, 11 tests). Every knowledge-base entry returned all required schema fields, with none missing, malformed or partially populated. Two edge cases confirmed input validation: differently cased variants of a valid dataset name resolved correctly, and an unrecognised name raised a clear error listing all valid options rather than failing silently.

test_score_explainer.py (Tier 1, 10 tests). In methodology mode, where no scoring API is available, all numeric fields correctly returned null. In API mode, fields populated correctly against a mocked payload. A deliberately fabricated gene identifier still returned a methodology explanation rather than being rejected, confirming that identifier validation is not

this tool’s responsibility. A regression guard inspects the module source directly for hardcoded numeric scores, confirming all scores are computed dynamically.

test_gene_alias.py (Tiers 1 and 3, 10 tests). Six mocked unit tests and four live tests exercising real network calls. Live tests passed across three scenarios: correct resolution with Ensembl and HGNC queried jointly; correct resolution when an Ensembl identifier was supplied directly; and graceful degradation when one source was unavailable. ERBB2 resolved correctly via HER2, confirming synonym handling against live production data. The failing case: with both sources simulated as unreachable, the local fallback was expected to execute but did not; root-cause analysis attributes this to an empty local index. This module spanning two tiers is recorded as an open structural inconsistency in the test suite.

test_api.py (Tier 2, 15 tests). Correct HTTP status codes across all tested failure modes: 422 for invalid input; 503 with retry headers for rate limiting from both inference providers; and a generic 500 for unexpected internal errors, with no internal detail leaked to the client. Score-explainer JSON parsed correctly into the frontend’s structured methodology object, robustly even when embedded within reasoning text or wrapped in code fences, falling back to plain prose where the output contract was not met.

test_latency.py (Tier 2, 8 tests). Response-time budgets met for both time-to-first-byte and overall data arrival. Where the language-model call itself fails, already-fetched tool data is still returned rather than failing the whole request.

test_tool.py (Tier 3, 1 test). Confirms the function-calling mechanism resolves correctly against a trivial test tool using a genuine, non-mocked model call a baseline proof of the mechanism before it is trusted against the three business tools.

test_agent_lambda.py (Tier 3, 6 tests). Executed directly against the deployed production function. Each of the three business-case tools was correctly selected across three distinct natural-language phrasings, which establishes routing over a small set of phrasings rather than general robustness to paraphrase. A fabricated gene–cell-line pair produced a structured methodology explanation rather than a fabricated numeric score. Two contrasting response-quality cases: for TP53 the model correctly elaborated beyond tool output, expanding LFS1 to Li-Fraumeni Syndrome 1; for a DepMap provenance query it asserted “three RNA expression sources”, a statement true of the corpus but not supported by the tool output it purported to describe. No test attempts to induce the agent to alter or contradict a ranking, so the architectural boundary is verified by construction rather than adversarially.

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